GSE 518 • PROBABILITY AND STATISTICS FOR ECONOMETRICS

Cal Poly MSQE • Prof. Carlos A. Flores

The Midterm Survival Guide

Everything from Set Theory through Multiple Random Variables, rebuilt from the ground up, in plain language.

SESSION 1 • 3 hours • target: 80%

SESSION 2 • 5 hours • target: 100%

CHEAT SHEET • 2 pages • bring it in

What this covers

Notes I through IX: Set Theory, Probability Theory, Conditional Probability and Independence, Random Variables, Distribution Functions, Transformations, Expectations and MGFs, Commonly Used Distributions, and Multiple Random Variables. Problem Sets 1 through 6. Per the syllabus, the midterm covers **Probability Theory only**, so everything from "Statistics" onward (random samples, estimation, asymptotics) is deliberately excluded.

Read this page first

You said the lectures turned into a jumble and that you freeze when you look at a problem because you cannot remember what the symbols mean. That is a *notation* problem wearing a *math* problem's clothing, and it is very fixable. This guide is built around that diagnosis.

> **The core claim of this guide:** this exam is not testing whether you can invent mathematics. It is testing whether you can (1) recognize which of about twelve situations you are in, and (2) execute the matching recipe without dropping a step. Dr. Flores said this himself: *"using formulas and processes right is more important than algebraic correctness."* That is enormously good news. It means partial credit lives in the setup, not the arithmetic.

How the two sessions differ

SESSION 1 **3 hours, target 80%**	SESSION 2 **5 hours, target 100%**
The five machines that generate roughly eight of the ten exam questions. Symbol decoding, expectation and variance, transformations, the joint-distribution machine, and covariance formulas.	Everything else, plus depth. Sets and axioms, Bayes, the distribution zoo, MGF technique, inequalities, the bivariate normal, and the traps that cost points under strict grading.
Aim: never be confused about *what to do*.	Aim: never be wrong about *how to do it*.
Ends with a 12-question checkpoint.	Ends with a full 10-question mock midterm plus four harder bonus problems.

The Sample Midterm is the Rosetta Stone

Dr. Flores's "Sample Questions from Old Midterms" sheet is in your project, and it maps one-to-one onto his notes about what appears. Every single question type he listed has a concrete example on that sheet. **Session 2 works all six of them completely**, and then gives you a fresh 10-question mock built on the same skeleton with different numbers.

His own framing on that sheet: about **3 to 4 "Easy" questions, 2 to 3 "Medium", and 1 to 2 "Hard"**, except he explicitly says *"For our class I do not plan to have Hard questions."* So budget your preparation accordingly: Easy and Medium are where all your points are.

Also from that sheet, his literal instruction: *"This is a closed-book exam. Please **show your work!** Try to show as much work and logic as possible."* Combined with his note that process beats algebra, the strategy is settled. Write more, not less.

Three rules for using this document

1. **Do not read passively.** Every **worked example** box is written so you can cover the solution, try it, then uncover. Reading a solved integral teaches you almost nothing. Failing at it for ninety seconds first teaches you a lot.
2. **The decoder boxes are the highest-value content in here.** If you only have one hour, read every decoder box and the cheat sheet.
3. **Build your own cheat sheet by hand.** Copy mine, but write it out. The act of transcribing is a third

of the studying. Also, you will find you can compress it further once you understand it, and that compression *is* the understanding.

1 The exam blueprint

This is reconstructed directly from Dr. Flores's own notes on the midterm. Read it as a map of where the points are.

Structure

9 to 10 questions, 10 to 11 points total, with multi-part questions labeled (a) through (e). One double-sided page of notes is allowed. Covers Probability Theory only, Problem Sets 1 through 6.

Question type	Weight	Where it lives in this guide
Joint distributions, the PS6 #2 machine: joint pdf → marginals → independence → conditional → conditional expectation → LIE	**4 to 5 of 10**	Session 1, Block 4 (full walkthrough of every part a through j)
Transformation of random variables, $Y = g(X)$, find f_Y	1 to 2	Session 1, Block 3
Given $X \sim N(\mu, \sigma^2)$, find the distribution of Y	1	Session 1, Block 3.5
Compute $\mathbb{E}[X]$ or $\mathrm{Var}[X]$ from a pdf/pmf	1	Session 1, Block 2
Covariance and correlation formulas: $\mathrm{Var}(aX + bY)$, $\mathrm{Cov}(aX + bY, cU + dV)$, "what is $\mathrm{Corr}(W, U)$?"	1 to 2	Session 1, Block 5
Alternates that may appear: Bayes' rule, Factorization Theorem	0 to 2	Session 2, Blocks 7 and 8

The Sample Midterm, mapped

Dr. Flores's notes name specific Sample Midterm questions. Here is the decoding:

SM #	Level	What it is	Twin in the problem sets
1	Easy	$X \sim N(1, 4)$, $Y = 2X - 1$, give the distribution of Y	PS3 #4
2	Easy	Joint pdf $e^{-(x_1+x_2+x_3)}$: are they independent?	PS6 #1(b)
3	Easy	Show $\mathrm{Var}(X) = p(1-p)$ for a Bernoulli	Notes VII, Example 1
4	Medium	Density of $Y = X^\beta$ from a Weibull density	PS3 #4, #5, #6
5	Medium	$f(x, y) = x + y$ on the unit square, four parts	**PS6 #2 (same machine)**
6	Hard	Show $\mathbb{E}[Y\|X] = \mathbb{E}\{\mathbb{E}[Y\|X, Z]\|X\}$	Notes IX, §6.1

Notice what just happened. Sample Midterm #5 and Problem Set 6 #2 are *the same problem with different numbers*. Both are: joint density on a rectangle, get the marginal, get the conditional, get

the conditional expectation, get a correlation. Dr. Flores said PS6 #2 supplies four to five of the ten questions and that SM #5 is a standard question. They are the same machine, and it runs half the exam.

Charming detail that will help you check your work: both problems have $\mathbb{E}[X] = \frac{7}{12}$.

The single most important sentence in Dr. Flores's notes

"Using formulas and processes right is more important than algebraic correctness."

Translation into exam strategy: write the formula down before you compute anything. Write the integral with its limits before you evaluate it. Name the theorem you are using. If you run out of time or botch the arithmetic, a correct, fully labeled setup is worth most of the point. A correct number with no visible reasoning is worth less than you think.

The four sentences you should be able to say instantly

If you can produce these four from memory under pressure, you have the spine of the exam:

1. "Marginal of X means integrate the joint over y, across the range y can take."
2. "Conditional equals joint over marginal, and I integrate over the *random* one."
3. "To transform, I find the support, check monotonicity, invert, and multiply by the absolute value of the derivative of the inverse."
4. "$\text{Var}(aX + bY) = a^2\text{Var}(X) + b^2\text{Var}(Y) + 2ab\text{Cov}(X, Y)$, and covariance is bilinear so I can expand it term by term."

Contents

Read this page first 2

1 The exam blueprint 4

2 Block 0. The symbol decoder 8

3 Block 0.5. Reading the question: signal to move 11

4 Block 1. The four descriptions of a random variable 13

5 Block 2. Expectation and variance 15

6 Block 3. Transformations of random variables 18

7 Block 3.5. The normal shortcut (a guaranteed question) 21

8 Block 4. The joint distribution machine 23

9 Block 5. Covariance, correlation, and variance of sums 29

10 Session 1 checkpoint 32

11 Block 6. Sets, axioms, and the proof toolkit 34

12 Block 7. Conditional probability, Bayes, and independence 37

13 Block 8. The distribution zoo 41

14 Block 9. Moment generating functions 44

15 Block 10. Markov, Chebyshev, Jensen 47

16 Block 11. The bivariate normal and iterated expectations 49

17 Block 12. The strict-grading checklist 51

18 Block 13. The real Sample Midterm, fully solved 53

19 Block 14. Mock midterm 57

20 Bonus: four harder problems 62

SESSION ONE

Three hours to eighty percent

20 min	Block 0	The symbol decoder
10 min	Block 0.5	Reading the question
25 min	Block 1	The four descriptions of a random variable
30 min	Block 2	Expectation and variance
35 min	Block 3	Transformations
45 min	Block 4	The joint distribution machine
25 min	Block 5	Covariance, correlation, variance of sums

2 Block 0. The symbol decoder

20 minutes

You told me you stare at problems and forget what μ and σ^2 mean. So we start there, and we start with the single most important distinction in the entire course.

2.1 Capital letters versus lowercase letters

The one that unlocks everything

X **(capital) is the random variable.** It is a *function*, a machine that has not run yet. Formally it maps outcomes to numbers. Informally: "the height of a randomly chosen student," before you have chosen anybody.

x **(lowercase) is a specific number it might equal.** A slot, a value, an input you are plugging in. "5.9 feet."

So $f_X(x)$ reads: *"the density function belonging to the random variable X, evaluated at the number x."* The subscript tells you *whose* function it is. The parenthesis tells you *where* you are evaluating it.

This matters because **you integrate or sum over the lowercase one, and the capital one is what the answer is about.** When you see $\int y\, f_{Y|X}(y|x)\, dy$, the y is being swept away by the integral and the x survives into the answer. That is why $\mathbb{E}[Y|X=x]$ comes out as a function of x.

Test yourself right now

Why is $\mathbb{E}[X]$ a *number* but $\mathbb{E}[Y|X]$ is a *random variable*?
Answer: because $\mathbb{E}[X]$ integrates x away and nothing is left. $\mathbb{E}[Y|X]$ integrates y away, but X is still sitting there, still random. This exact distinction is what makes the Law of Iterated Expectations work, and it is the punchline of PS6 #2(j).

2.2 The full symbol table

Read this once slowly. Then come back to it every time you get stuck.

Symbol	Say it out loud as	What it actually is
Ω or S	"the sample space"	The set of every possible outcome. Flip two coins: $\{HH, HT, TH, TT\}$.
A^c, $\bar{A}$	"not A"	Everything in Ω outside A.
$A \cap B$	"A and B"	Both happen.
$A \cup B$	"A or B"	At least one happens.
$\emptyset$	"the empty set"	Nothing. $P(\emptyset) = 0$.
$\mathcal{B}$	"the Borel field"	The list of sets you are allowed to assign probability to. On the exam: ignore it.
X	"big X"	The random variable. A function from outcomes to numbers.
x	"little x"	A particular value X could take.

Symbol	Say it out loud as	What it actually is
$\mathcal{X}$	"the support of X"	The set of x where $f_X(x) > 0$. **The single most commonly dropped item on this exam.**
$X \sim$	"X is distributed as"	"$X \sim N(0,1)$" means X follows that distribution.
$F_X(x)$	"the cdf"	$\mathrm{P}(X \leq x)$. Always between 0 and 1. Always defined for every real x.
$f_X(x)$	"the pdf" or "the pmf"	Discrete: $\mathrm{P}(X = x)$, an actual probability. Continuous: a density, *not* a probability, and it can exceed 1.
$M_X(t)$	"the mgf"	$\mathbb{E}[e^{tX}]$. A fingerprint of the distribution and a moment-vending machine.
$\mathbb{E}[X]$, μ	"the expectation" or "the mean"	The probability-weighted average of X. A single number, the center.
μ_X	"mu sub X"	Just $\mathbb{E}[X]$, written short.
$\mathrm{Var}(X)$, σ^2	"the variance"	$\mathbb{E}[(X-\mu)^2]$. Average squared distance from the mean. Spread. In squared units.
σ, σ_X	"sigma", "the standard deviation"	$\sqrt{\mathrm{Var}(X)}$. Spread in the *original* units.
σ_{XY}	"sigma X Y"	Shorthand for $\mathrm{Cov}(X,Y)$.
ρ, ρ_{XY}	"rho"	The correlation, $\mathrm{Cov}(X,Y)/(\sigma_X\sigma_Y)$. Always in $[-1,1]$. Unitless.
λ	"lambda"	Parameter of the Exponential and Poisson.
p		Probability of "success". Bernoulli, Binomial, Geometric.
n		Number of trials.
α, β	"alpha, beta"	Shape and scale parameters, Gamma and Beta.
$\Gamma(\alpha)$	"the gamma function"	A continuous version of the factorial. $\Gamma(n) = (n-1)!$
$\Phi(z)$	"big phi"	The *cdf* of the standard normal. What the z-table gives you.
$\phi(z)$	"little phi"	The *pdf* of the standard normal, the bell curve itself.
$\chi^2_{(p)}$	"chi-squared with p degrees of freedom"	A distribution. p is its only parameter.
$g(X)$	"g of X"	Any function you apply to X, e.g. X^2, $\ln X$, $2X+3$.
$g^{-1}(y)$	"g inverse of y"	Solve $y = g(x)$ for x. That is all this ever means.
$\left\lvert \frac{\mathrm{d}g^{-1}(y)}{\mathrm{d}y} \right\rvert$	"the Jacobian"	Take that solved-for x, differentiate with respect to y, take absolute value.
$f_{XY}(x,y)$	"the joint"	Probability/density that X and Y land at x and y together.
$f_{Y\|X}(y\|x)$	"the conditional of Y given X"	The distribution of Y once you already know $X = x$.
$\mathbb{E}[Y\|X=x]$	"the conditional expectation"	The mean of Y inside the slice where $X = x$.
$\perp\!\!\!\perp$	"independent"	$f_{XY} = f_X \cdot f_Y$.

Two Greek letters people constantly mix up

$\boldsymbol{\mu}$ is a *location*: slide the whole distribution left or right. $\boldsymbol{\sigma^2}$ is a *spread*: squeeze or stretch it. In $N(\mu, \sigma^2)$ the two slots are always **(center, spread-squared)**. So $N(5, 16)$ means centered at 5, standard deviation 4, not 16.

Watch for this on the exam. If a question says "$X \sim N(1, 4)$", then $\sigma = 2$, and standardizing means dividing by 2, not by 4. PS5 #1 turns exactly on this.

3 Block 0.5. Reading the question: signal to move

10 minutes

This is the table you actually want taped inside your head. Almost every question on this midterm announces itself with a phrase. Learn the phrase, learn the move.

When the problem says…	Your first move is…
"Find the value of c that makes this a pdf"	Set $\int_{\text{support}} f = 1$ and solve for c. (Discrete: set the sum to 1.)
"Find the cdf $F_X(x)$"	$F_X(x) = \int_{-\infty}^{x} f_X(r)\,\mathrm{d}r$. Then *write it piecewise over all of* $\mathbb{R}$, including the 0 branch and the 1 branch.
"Find $\mathrm{P}(a \le X \le b)$"	$F(b) - F(a)$, or integrate the pdf from a to b. Same thing.
"Is this a valid pdf / cdf / pmf?"	Check the two (pdf/pmf) or three (cdf) properties in the checklist. Say their names.
"$Y = g(X)$. Find the pdf of Y"	The four-step change of variable. Support, monotone, invert, Jacobian.
"$X \sim N(\mu, \sigma^2)$ and $Y = aX + b$"	Instantly: $Y \sim N(a\mu + b,\ a^2\sigma^2)$. Then standardize if a probability is wanted.
"Find $\mathbb{E}[X]$"	$\int x f_X(x)\,\mathrm{d}x$ over the support.
"Find $\mathrm{Var}(X)$"	$\mathbb{E}[X^2] - (\mathbb{E}[X])^2$. Almost never the definition directly.
"Find the marginal $f_X(x)$"	Integrate the joint over the *other* variable, y, across everything y can be.
"Are X and Y independent?"	Factorization Theorem. Can you write $f_{XY}(x,y) = g(x)h(y)$? Also check the support is a rectangle.
"Find $f_{Y\|X}(y\|x)$"	Joint divided by the marginal of the *conditioning* variable: f_{XY}/f_X.
"Find $\mathbb{E}[Y\|X = x]$"	$\int y\, f_{Y\|X}(y\|x)\,\mathrm{d}y$, integrating over y only. Answer is a function of x.
"Find $\mathbb{E}[\,\mathbb{E}(Y\|X)\,]$"	Law of Iterated Expectations. It equals $\mathbb{E}[Y]$. Do not compute the integral.
"What is $\mathrm{Corr}(W, U)$ where W, U are linear combinations?"	Bilinearity of covariance, then $\mathrm{Var}(aX + bY)$, then divide. Pure formula work.
"Show that X follows a ___ distribution"	MGF. Compute $M_X(t)$, match it to a known one, cite Theorem 2.3.11.
"Find a bound on $\mathrm{P}(\cdot)$"	Markov if $X \ge 0$; Chebyshev if it is about distance from the mean.
"Compare $\mathbb{E}[g(X)]$ to $g(\mathbb{E}[X])$"	Jensen. Convex gives $\ge$, concave gives $\le$.
"You are given $\mathrm{P}(B\|A)$ but asked for $\mathrm{P}(A\|B)$"	Bayes' rule. The denominator is the Law of Total Probability.
"A partition… then for any event B"	Law of Total Probability: $\mathrm{P}(B) = \sum_i \mathrm{P}(B\|A_i)\mathrm{P}(A_i)$.

If you are truly stuck on any problem, run this in order

1. **Write down the support.** What values can this variable take? Half the time this reveals the answer.

2. **Name the object you are asked for.** Is it a cdf, a pdf, a number, a distribution?
3. **Name the object you were given.** Same list.
4. **Find the bridge.** There is a small, fixed set of bridges between those objects (see the map on the next page).
5. **Write the formula with limits before computing anything.** You have now earned most of the point.

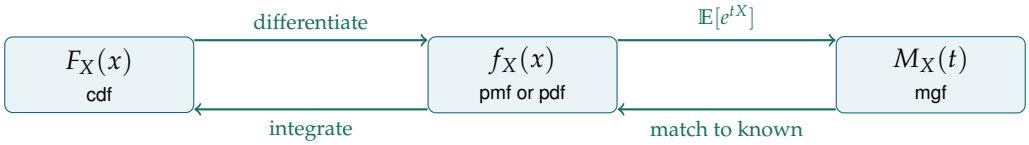
4 Block 1. The four descriptions of a random variable

25 minutes

4.1 What a random variable actually is

A random variable is a **function that turns outcomes into numbers**. Flip two coins; the outcome is HT, which is not a number. Define $X =$ number of heads, and now $HT \mapsto 1$. That is the whole idea. We do this because numbers can be integrated, graphed, averaged, and sets cannot.

Once X exists, we describe it in one of four equivalent ways. Any one of them pins down the distribution completely.



4.2 The cdf: $F_X(x) = \mathrm{P}(X \le x)$

How to think about the cdf

It is a running total. Walk along the number line from $-\infty$ to the right, and $F_X(x)$ is how much probability you have swept up by the time you reach x. It starts at 0, ends at 1, and never goes down.

For a **discrete** variable it is a staircase, and the height of each jump is the probability at that point. For a **continuous** variable it is a smooth curve.

Validity checklist: is F_X a legitimate cdf?

A function is a cdf if and only if all three hold:

1. $\lim_{x\to-\infty} F_X(x) = 0$ and $\lim_{x\to\infty} F_X(x) = 1$
2. Non-decreasing: $x_1 \le x_2 \Rightarrow F_X(x_1) \le F_X(x_2)$. (Show $F' \ge 0$.)
3. Right continuous: $\lim_{x\to x_0^+} F_X(x) = F_X(x_0)$.

4.3 The pmf and the pdf

The distinction that trips everyone

pmf (discrete): $f_X(x) = \mathrm{P}(X = x)$. This *is* a probability. It lives in $[0, 1]$.

pdf (continuous): $f_X(x)$ is a *density*, not a probability. It can be bigger than 1. Only its *area* is a probability. For a continuous variable $\mathrm{P}(X = x) = 0$ for every single point, which is why $\mathrm{P}(a < X < b) = \mathrm{P}(a \le X \le b)$ in the continuous case only.

PS3 #1 asks exactly this. Answer: no, a pdf need not be below 1. Counterexample: $X \sim \text{Uniform}(0, \frac{1}{2})$ has $f_X(x) = 2$.

Validity checklist: is f_X a legitimate pmf or pdf?

Only two conditions:

1. $f_X(x) \geq 0$ for all $x \in \mathbb{R}$
2. $\sum_x f_X(x) = 1$ (pmf) or $\int_{-\infty}^{\infty} f_X(x)\,dx = 1$ (pdf)

Worked: PS3 #3, the classic "find c, then find the cdf"

Let $f_X(x) = \begin{cases} 0.2 & -1 \leq x \leq 0 \\ 0.2 + cx & 0 < x \leq 1 \end{cases}$

(a) Find c. Signal: "makes it a pdf" means set the total area to 1. Break the integral at the kink.

$$\int_{-1}^{0} 0.2\,dx + \int_{0}^{1} (0.2 + cx)\,dx = 0.2 + \left(0.2x + \tfrac{c}{2}x^2\right)\Big|_0^1 = 0.2 + 0.2 + \tfrac{c}{2} = 1$$

So $\frac{c}{2} = 0.6$ and $\boxed{c = 1.2}$.

(b) Find the cdf. $F_X(x) = \int_{-\infty}^{x} f$. Handle each piece separately.

For $-1 \leq x \leq 0$: $F_X(x) = \int_{-1}^{x} 0.2\,dr = 0.2x + 0.2$.

For $0 < x \leq 1$: $F_X(x) = \underbrace{0.2}_{\text{all of the first piece}} + \int_0^x (0.2 + 1.2r)\,dr = 0.2 + 0.2x + 0.6x^2$.

$$F_X(x) = \begin{cases} 0 & x < -1 \\ 0.2x + 0.2 & -1 \leq x \leq 0 \\ 0.2 + 0.2x + 0.6x^2 & 0 < x \leq 1 \\ 1 & x > 1 \end{cases}$$

(c) $P(0 \leq X \leq 0.5) = F(0.5) - F(0) = [0.2 + 0.1 + 0.6(0.25)] - 0.2 = 0.45 - 0.2 = \boxed{0.25}$

Where points get lost on cdf questions

The top and bottom rows. Students write only the two middle branches and lose credit. **A cdf must be defined for every real number.** Always add "0 if $x <$ (left end)" and "1 if $x >$ (right end)". This is free money and it is the most common deduction in this course.

5 Block 2. Expectation and variance

30 minutes

Dr. Flores's notes list "calculating $\mathbb{E}[X]$ or $\mathrm{Var}[X]$" as a standard exam question. It is one of the most reliable points on the paper, so do not fumble it.

5.1 The definitions, and why they look like that

Expectation is a weighted average

$$\mathbb{E}[X] = \begin{cases} \displaystyle\sum_{x \in \mathcal{X}} x\, f_X(x) & \text{discrete} \\ \displaystyle\int_{x \in \mathcal{X}} x\, f_X(x)\, \mathrm{d}x & \text{continuous} \end{cases}$$

Read it as: *"take each value x, weight it by how likely that value is, add it all up."* That is genuinely all it is. The integral sign is doing the job that "add it all up" does in the discrete case.

$\mathcal{X}$ under the integral means **integrate over the support only**, the region where $f_X > 0$. If $f_X(x) = 2/x^3$ for $x > 1$, your limits are 1 to ∞, not $-\infty$ to ∞.

LOTUS: expectation of a function

$$\mathbb{E}[g(X)] = \int_{x \in \mathcal{X}} g(x) f_X(x)\, \mathrm{d}x$$

You do not need the distribution of $g(X)$. You just slide g in where x used to be and keep the original density. So $\mathbb{E}[X^2] = \int x^2 f_X(x)\, \mathrm{d}x$. This saves enormous time and is used constantly.

5.2 The properties you will actually use

Linearity (the workhorse). For constants a, b:

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b \qquad \mathbb{E}[c] = c$$

Non-linearity (the trap). In general $\mathbb{E}[g(X)] \neq g(\mathbb{E}[X])$. In particular

$$\mathbb{E}[X^2] \neq (\mathbb{E}[X])^2, \qquad \mathbb{E}[e^X] \neq e^{\mathbb{E}[X]}, \qquad \mathbb{E}[1/X] \neq 1/\mathbb{E}[X]$$

The gap between $\mathbb{E}[X^2]$ and $(\mathbb{E}[X])^2$ has a name. It is the variance.

5.3 Variance: definition, shortcut, and scaling

What variance measures

$\mathrm{Var}(X) = \mathbb{E}[(X - \mu)^2]$ where $\mu = \mathbb{E}[X]$. In words: *"on average, how far squared is X from its own center?"* We square so that being below and above the mean do not cancel out. $\sigma = \sqrt{\mathrm{Var}(X)}$ is the standard deviation, and it is in the same units as X. Variance is in squared units, which is why

it is harder to interpret.

The three variance facts. Memorize these cold.

(V1) Computational shortcut:	$\mathrm{Var}(X) = \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2$
(V2) Scaling and shifting:	$\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X)$
(V3) Of a function:	$\mathrm{Var}\big(g(X)\big) = \mathbb{E}\big[g(X)^2\big] - \big(\mathbb{E}[g(X)]\big)^2$

(V1) is how you should compute variance essentially every time. Get $\mathbb{E}[X]$, get $\mathbb{E}[X^2]$, subtract the square of the first from the second. Going through the definition directly is slower and more error-prone.

(V2) says adding a constant does nothing to spread (sliding the whole distribution does not stretch it) **and multiplying scales the variance by** a^2 (because variance is in squared units). The absence of b on the right is not a typo.

Worked: PS4 #2, and the surprise ending

$f_X(x) = \dfrac{2}{x^3}$ for $x > 1$, zero otherwise. Support is $(1, \infty)$.

(a) Check it is a pdf.

$$\int_1^\infty \frac{2}{x^3}\,dx = \left[-\frac{1}{x^2}\right]_1^\infty = 0 - (-1) = 1 \quad \checkmark$$

(b) Find $\mathbb{E}[X]$**.**

$$\mathbb{E}[X] = \int_1^\infty x \cdot \frac{2}{x^3}\,dx = \int_1^\infty \frac{2}{x^2}\,dx = \left[-\frac{2}{x}\right]_1^\infty = 0 - (-2) = \boxed{2}$$

(c) Find $\mathrm{Var}(X)$**.** Use (V1), so first get $\mathbb{E}[X^2]$:

$$\mathbb{E}[X^2] = \int_1^\infty x^2 \cdot \frac{2}{x^3}\,dx = \int_1^\infty \frac{2}{x}\,dx = \Big[2\ln x\Big]_1^\infty = \infty$$

The integral diverges, so $\mathbb{E}[X^2]$ does not exist, so $\mathrm{Var}(X)$ **does not exist**.

"Does not exist" is a real answer

Expectations only exist if the integral or sum converges. A distribution can have a perfectly good mean and an infinite variance (heavy tails). The t-distribution with $r = 1$, the Cauchy, does not even have a mean. If an integral blows up, say so explicitly and name it. Do not write a number.

Worked: discrete uniform, PS4 #1(b) pattern

X uniform on $\{1, 2, \ldots, n\}$, so $f_X(x) = 1/n$ on that set. You need two series facts:

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}, \qquad \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\mathbb{E}[X] = \sum_{i=1}^{n} i \cdot \frac{1}{n} = \frac{1}{n} \cdot \frac{n(n+1)}{2} = \frac{n+1}{2}$$

$$\mathbb{E}[X^2] = \frac{1}{n} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{(n+1)(2n+1)}{6}$$

$$\mathrm{Var}(X) = \frac{(n+1)(2n+1)}{6} - \left(\frac{n+1}{2}\right)^2 = \frac{2n^2+3n+1}{6} - \frac{n^2+2n+1}{4} = \boxed{\frac{n^2-1}{12}}$$

Continuous uniform, worth memorizing: PS4 #4

$X \sim \mathrm{Uniform}(a,b)$, $f_X(x) = \dfrac{1}{b-a}$ on (a,b).

$$\mathbb{E}[X] = \frac{a+b}{2}, \qquad \mathbb{E}[X^2] = \frac{a^2+ab+b^2}{3}, \qquad \mathrm{Var}(X) = \frac{(b-a)^2}{12}$$

6 Block 3. Transformations of random variables

35 minutes

Dr. Flores flags this as a guaranteed question type (his notes cite Sample Midterm #4 and "a lot of" PS3). This is a pure recipe. Learn the four steps and this becomes free points.

6.1 The setup in plain language

You know everything about X. Someone hands you $Y = g(X)$, some function of it: $Y = 2X + 3$, $Y = X^2$, $Y = -\ln X$. Since X is random, Y is random too. **What is the distribution of Y?**

Why we care about "strictly monotone"

If g is strictly increasing or strictly decreasing (never both), then each y came from exactly one x, so you can run the machine backwards: $x = g^{-1}(y)$. Clean.

If g is not monotone, several x values map to the same y and you have to add up all the sources. Example: $Y = X^2$ with $X \in \mathbb{R}$. Then $y = 4$ could have come from $x = 2$ *or* $x = -2$, so $P(Y = 4)$ must combine both. Messy.

Crucial nuance that shows up in PS3 #6(c): g only has to be monotone *on the support of* X, not everywhere. $Y = X^2$ is not monotone on $\mathbb{R}$, but if X lives on $(0, 1)$ then on that interval it is strictly increasing and the recipe works fine. Always check monotonicity *over the support*, and say that you did.

6.2 The four-step recipe (continuous case)

Change of variable formula, Result 3 / Theorem 2.1.5

Step 1. Find the support of Y. Push the endpoints of $\mathcal{X}$ through g. If $x \in (0, 1)$ and $y = -\ln x$, then as $x \to 0^+$, $y \to \infty$; at $x = 1$, $y = 0$. So $\mathcal{Y} = (0, \infty)$.

Step 2. Check strict monotonicity on $\mathcal{X}$. Compute $\dfrac{dg(x)}{dx}$ and show it keeps one sign on the support. Write the sentence.

Step 3. Invert. Solve $y = g(x)$ for x, giving $x = g^{-1}(y)$. Then differentiate it: $\dfrac{dg^{-1}(y)}{dy}$.

Step 4. Apply the formula.

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \left| \frac{dg^{-1}(y)}{dy} \right| \quad \text{for } y \in \mathcal{Y}, \qquad 0 \text{ otherwise}$$

The three ways people lose points here

1. **Forgetting the absolute value.** If g is decreasing, the derivative is negative and you get a negative "density", which is nonsense. The bars are not decoration.
2. **Forgetting to state the new support.** A pdf without its range is an incomplete answer. Always finish with "for $y \in \ldots$, and zero otherwise."

3. **Plugging y into f_X instead of $g^{-1}(y)$.** You feed f_X an x-value. That x-value happens to be written as $g^{-1}(y)$. Substitute carefully.

Worked: the canonical example, $X \sim$ Uniform$(0,1)$, $Y = -\ln X$

Here $f_X(x) = 1$ on $(0,1)$ and $F_X(x) = x$.
Step 1. $x \in (0,1) \Rightarrow y = -\ln x \in (0,\infty)$. So $\mathcal{Y} = (0,\infty)$.
Step 2. $\dfrac{dy}{dx} = -\dfrac{1}{x} < 0$ for all $x \in (0,1)$, so strictly decreasing. ✓
Step 3. $y = -\ln x \Rightarrow -y = \ln x \Rightarrow x = g^{-1}(y) = e^{-y}$, and $\dfrac{dg^{-1}}{dy} = -e^{-y}$.
Step 4.

$$f_Y(y) = f_X(e^{-y}) \cdot \left|-e^{-y}\right| = (1) \cdot e^{-y} = e^{-y}, \qquad y \in (0,\infty)$$

So $Y \sim$ Exponential with the professor's parameter $\lambda = 1$.
Bonus, the cdf route (Result 2). Since g is decreasing, $F_Y(y) = 1 - F_X(g^{-1}(y)) = 1 - e^{-y}$. Differentiate and you get e^{-y} again. Two roads, same destination.

Result 2: the cdf version, for when you are asked for F_Y directly

$$g \text{ strictly } \textbf{increasing}: \quad F_Y(y) = F_X\big(g^{-1}(y)\big)$$

$$g \text{ strictly } \textbf{decreasing}: \quad F_Y(y) = 1 - F_X\big(g^{-1}(y)\big)$$

The "$1-$" appears because flipping the inequality direction turns $\mathrm{P}(X \le \cdot)$ into $\mathrm{P}(X \ge \cdot)$, and $\mathrm{P}(A) = 1 - \mathrm{P}(A^c)$.

6.3 The discrete case: same recipe, one term deleted

Result 4, discrete transformations

$$f_Y(y) = f_X\big(g^{-1}(y)\big) \quad \text{for } y \in \mathcal{Y}, \qquad 0 \text{ otherwise}$$

There is no Jacobian in the discrete case. No absolute value, no derivative. You are just relabeling which outcome carries which probability, and probabilities do not get stretched by relabeling. Densities do, which is why the continuous case needs the correction factor.

Worked: PS3 #5, discrete with an ugly-looking support

$X \in \{0,1,2,\dots\}$ with $f_X(x) = \left(\frac{1}{3}\right)\left(\frac{2}{3}\right)^x$. Let $Y = \dfrac{X}{X+1}$.
Step 1, support. $x = 0 \Rightarrow y = 0$; $x = 1 \Rightarrow y = \frac{1}{2}$; $x = 2 \Rightarrow y = \frac{2}{3}$; $x = 3 \Rightarrow y = \frac{3}{4}$. So $\mathcal{Y} = \left\{0, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots\right\} = \left\{\frac{x}{x+1}\right\}$.
Step 2, monotone. $\dfrac{dy}{dx} = \dfrac{1}{(x+1)^2} > 0$. Strictly increasing. ✓
Step 3, invert. $y = \dfrac{x}{x+1} \Rightarrow y(x+1) = x \Rightarrow x = \dfrac{y}{1-y}$.

Step 4. No Jacobian:

$$f_Y(y) = \left(\tfrac{1}{3}\right)\left(\tfrac{2}{3}\right)^{\frac{y}{1-y}} \quad \text{for } y \in \left\{0, \tfrac{1}{2}, \tfrac{2}{3}, \tfrac{3}{4}, \dots\right\}, \quad 0 \text{ otherwise}$$

Worked: PS3 #6(b) and (c), two fast ones

(b) $f_X(x) = 7e^{-7x}$ on $(0,\infty)$, $Y = 4X + 3$.

- Support: $x \in (0,\infty) \Rightarrow y \in (3,\infty)$.
- $\mathrm{d}y/\,\mathrm{d}x = 4 > 0$, increasing. ✓
- $x = g^{-1}(y) = \dfrac{y-3}{4}$, so $\dfrac{\mathrm{d}g^{-1}}{\mathrm{d}y} = \dfrac{1}{4}$.
- $f_Y(y) = 7e^{-7(y-3)/4} \cdot \dfrac{1}{4} = \dfrac{7}{4}e^{-7(y-3)/4}$ for $3 < y < \infty$.

(c) $f_X(x) = 30x^2(1-x)^2$ on $(0,1)$, $Y = X^2$.

- Support: $x \in (0,1) \Rightarrow y \in (0,1)$.
- $\mathrm{d}y/\,\mathrm{d}x = 2x > 0$ *on* $(0,1)$. Note it is not monotone on all of $\mathbb{R}$, but we only need the support. Say this out loud in your answer.
- $x = g^{-1}(y) = \sqrt{y}$, so $\dfrac{\mathrm{d}g^{-1}}{\mathrm{d}y} = \tfrac{1}{2}y^{-1/2}$.
- $f_Y(y) = 30\,y\,(1-\sqrt{y})^2 \cdot \tfrac{1}{2}y^{-1/2} = 15\,y^{1/2}(1-y^{1/2})^2$ for $0 < y < 1$.

Sanity check: $\int_0^1 15y^{1/2} - 30y + 15y^{3/2}\,\mathrm{d}y = 10 - 15 + 6 = 1$. ✓

7 Block 3.5. The normal shortcut (a guaranteed question)

included above

Dr. Flores's note says an "applications" question appears: *given* $X \sim N(\ldots)$, *find the distribution of* Y. This is PS3 #4, and it is worth having as an instant reflex.

The result. If $X \sim N(\mu, \sigma^2)$ and $Y = aX + b$ with $a \neq 0$, then

$$\boxed{Y \sim N\left(a\mu + b,\ a^2\sigma^2\right)}$$

Why those two numbers? Straight from Block 2: $\mathbb{E}[aX + b] = a\mu + b$ by linearity, and $\text{Var}(aX + b) = a^2\sigma^2$ by (V2). The only genuinely new content is that the *shape* stays normal.

The special case that runs the whole exam. Set $a = 1/\sigma$ and $b = -\mu/\sigma$:

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

This is called standardizing, and it is how you compute any normal probability.

Deriving it from the pdf, if the exam asks you to "show" it

This is PS3 #4 done with the four-step recipe.

1. Support of X is $\mathbb{R}$, so support of $Y = aX + b$ is $\mathbb{R}$.
2. $\mathrm{d}y/\,\mathrm{d}x = a \neq 0$, so strictly monotone either way. ✓
3. $x = g^{-1}(y) = \dfrac{y - b}{a}$, so $\dfrac{\mathrm{d}g^{-1}}{\mathrm{d}y} = \dfrac{1}{a}$.
4. Apply the formula:

$$f_Y(y) = \frac{1}{|a|} \cdot \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{ -\frac{\left(\frac{y-b}{a} - \mu\right)^2}{2\sigma^2} \right\} = \frac{1}{\sqrt{2\pi(a\sigma)^2}} \exp\left\{ -\frac{\big(y - (a\mu + b)\big)^2}{2(a^2\sigma^2)} \right\}$$

The last step uses $|a| = \sqrt{a^2}$ to pull $|a|$ inside the square root, and factors a^2 out of the squared bracket. Recognize the result: it is exactly the $N(a\mu + b,\ a^2\sigma^2)$ density.

Worked: PS5 #1, standardizing to get a number

$X \sim N(\mu = 1, \sigma^2 = 4)$, so $\sigma = 2$. Find $\mathrm{P}(X < 1)$ and $\mathrm{P}(X \geq 1.5)$.

$\mathrm{P}(X < 1)$**:** standardize. $\mathrm{P}\left(\dfrac{X - 1}{2} < \dfrac{1 - 1}{2}\right) = \mathrm{P}(Z < 0) = \boxed{0.5}$ by symmetry, no table needed.

$\mathrm{P}(X \geq 1.5)$**:**

$$\mathrm{P}(X \geq 1.5) = 1 - \mathrm{P}(X < 1.5) = 1 - \mathrm{P}\left(Z < \frac{1.5 - 1}{2}\right) = 1 - \mathrm{P}(Z < 0.25) = 1 - \Phi(0.25)$$

From the table $\Phi(0.25) = 0.5987$, so the answer is $\boxed{0.4013}$.

Divide by σ, not σ^2

The variance was 4, so you divide by $\sqrt{4} = 2$. Dividing by 4 is the single most common error on normal problems. Whenever you see $N(\mu, \sigma^2)$, immediately write down $\sigma = \sqrt{\sigma^2}$ in the margin before doing anything else.

Standard normal facts you should not have to look up

- $\phi(z) = \phi(-z)$: the bell curve is symmetric.
- $\Phi(-z) = 1 - \Phi(z)$: use this when your table only has positive z.
- $\Phi(0) = 0.5$.
- If $Z \sim N(0,1)$ then $Z^2 \sim \chi^2_{(1)}$ (PS5 #4b).
- There is **no closed form** for the normal cdf. That is why tables exist. If an exam question asks for a probability, the expected answer is either a $\Phi(\cdot)$ expression or a table lookup.

8 Block 4. The joint distribution machine

45 minutes

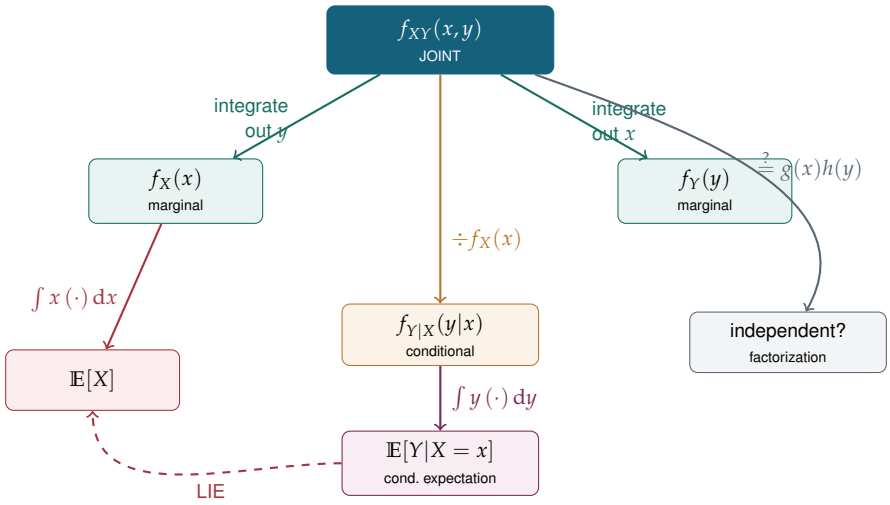
Read this before anything else in this block

Dr. Flores wrote: *"Problem Set 6 #2 parts a-j typically make up 4-5 questions on the 10 question exam."*

That is up to half your grade sitting inside one problem structure. If you master nothing else in this guide, master this block. Everything here is one integral repeated with different limits and different integrands, and once you see that, it stops being scary.

8.1 The whole block in one picture

Everything below is derived from one object, the joint density $f_{XY}(x, y)$. Here is the map:



8.2 The four definitions

Joint, marginal, conditional: what each one means in words

Joint $f_{XY}(x, y)$: how much probability mass sits at the *pair* (x, y). Think of a mound of dirt sitting over a region of the floor.

Marginal $f_X(x)$: ignore Y entirely, what is the distribution of X by itself? You get it by **integrating** y **away**: sweep up all the dirt in the column above each x value.

$$f_X(x) = \int_{y \in \mathcal{Y}} f_{XY}(x, y)\, dy \qquad f_Y(y) = \int_{x \in \mathcal{X}} f_{XY}(x, y)\, dx$$

Mnemonic: to get the marginal of X, integrate out the other one. The variable you integrate is the one that *disappears*. If your "marginal of X" still has a y in it, you integrated the wrong variable.

Conditional $f_{Y|X}(y|x)$: you have been told $X = x$. Slice the mound at that x, and rescale the slice so it integrates to 1 (because it has to be a legitimate density on its own).

$$f_{Y|X}(y|x) = \frac{f_{XY}(x, y)}{f_X(x)}$$

The denominator is precisely the rescaling factor. It is the marginal of the variable being *conditioned on*.

Conditional expectation $\mathbb{E}[Y|X=x]$: the average of Y within that slice.

$$\mathbb{E}[Y|X=x] = \int_{y\in\mathcal{Y}} y\, f_{Y|X}(y|x)\, dy$$

The question that decides everything: which variable is random?

In $f_{Y|X}(y|x)$, Y **is random and** x **is a fixed given number.** Therefore:

- You integrate over y, never over x.
- $\int_{\mathcal{Y}} f_{Y|X}(y|x)\, dy = 1$ ✓ but $\int_{\mathcal{X}} f_{Y|X}(y|x)\, dx \neq 1$ in general ×
- The answer to $\mathbb{E}[Y|X=x]$ is a *function of* x, and that is correct, not a mistake.

Dr. Flores flags this in the notes as the key note for the whole chapter. Every time you set up an integral in this block, write in the margin: *which one is random?*

8.3 Independence and the Factorization Theorem

Factorization Theorem (Lemma 4.2.7), flagged as a possible exam question

X and Y are independent **if and only if** you can write

$$f_{XY}(x,y) = g(x)\cdot h(y)$$

for some pair of functions, for all (x,y). The functions do *not* need to be the marginals themselves; any valid split works, constants can go anywhere.

How to actually use it in ten seconds: try to split the expression into an x-only chunk times a y-only chunk.

- $f_{XY} = e^{-x}e^{-y}$ ✓ splits. **Independent.**
- $f_{XY} = 0.25(2x+y) = 0.5x + 0.25y$ × A *sum* of x and y terms cannot be split into a product. **Not independent.**
- $f_{XY} = 6x^2y$ ✓ $= (6x^2)(y)$. **Independent.**

The hidden condition: the support must be a rectangle. If the region is like $0 < x < y < 1$, where the bounds of one variable involve the other, they are **not** independent even if the formula looks separable. Knowing X tells you something about the range of Y, and that is dependence.

What independence buys you, once you have it

- $f_{Y|X}(y|x) = f_Y(y)$ and $f_{X|Y}(x|y) = f_X(x)$. Conditioning tells you nothing.
- $\mathbb{E}[r(X)s(Y)] = \mathbb{E}[r(X)]\cdot\mathbb{E}[s(Y)]$ for any functions. Special case: $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.
- Therefore $\mathrm{Cov}(X,Y) = 0$ and $\rho = 0$.
- If $Z = X+Y$ then $M_Z(t) = M_X(t)\cdot M_Y(t)$ (Thm. 4.2.12). MGFs multiply.
- If $U = g(X)$ and $V = h(Y)$, then U and V are independent too (Thm. 4.3.5).
- $\mathrm{Var}(aX+bY) = a^2\mathrm{Var}(X) + b^2\mathrm{Var}(Y)$, the cross term vanishes.

The converse is false, and Dr. Flores hates this error

$$\text{Cov}(X, Y) = 0 \not\Rightarrow X \text{ and } Y \text{ independent}$$

Covariance only detects *linear* dependence. Two variables can be perfectly deterministically related and still have zero covariance (e.g. $Y = X^2$ with X symmetric about 0). The notes call the jump from "$\rho = 0$" to "X does not cause Y" the "worst jump" with three NOs. Expect a conceptual question on this.

The one exception: if (X, Y) is *bivariate normal*, then $\rho = 0$ *does* imply independence, in both directions. That is a special property of the bivariate normal, not a general fact.

8.4 The complete PS6 #2 walkthrough

Here is the problem Dr. Flores says generates four to five exam questions, worked end to end with nothing skipped. Cover the solutions and try each part first.

Setup. $f_{XY}(x,y) = 0.25(2x+y)$ for $0 < x < 1$, $0 < y < 2$, and zero otherwise.
Before touching anything: **write down the support**. x runs 0 to 1, y runs 0 to 2. Those are your integration limits for the rest of the problem, and mixing them up is the number one source of error here.

(a) Find $\mathrm{P}(X < 0.25,\ Y < 1)$

A joint probability over a region means a double integral of the joint density over that region. Intersect the requested region with the support.

$$\mathrm{P}(X < 0.25, Y < 1) = \int_0^1 \int_0^{0.25} 0.25(2x+y)\,dx\,dy$$

Inner integral in x (treat y as a constant):

$$\int_0^{1/4} (2x+y)\,dx = \Big[x^2 + xy\Big]_0^{1/4} = \frac{1}{16} + \frac{y}{4}$$

Outer integral in y:

$$0.25 \int_0^1 \left(\frac{1}{16} + \frac{y}{4}\right) dy = 0.25 \left[\frac{y}{16} + \frac{y^2}{8}\right]_0^1 = 0.25\left(\frac{1}{16} + \frac{1}{8}\right) = 0.25 \cdot \frac{3}{16} = \boxed{\frac{3}{64} \approx 0.047}$$

(b) Find the marginals $f_X(x)$ and $f_Y(y)$

Marginal of X: integrate y away, over $y \in (0,2)$.

$$f_X(x) = \int_0^2 0.25(2x+y)\,dy = 0.25\Big[2xy + \tfrac{y^2}{2}\Big]_0^2 = 0.25(4x+2) = \boxed{x + \tfrac{1}{2}}, \quad 0 < x < 1$$

Marginal of Y: integrate x away, over $x \in (0,1)$.

$$f_Y(y) = \int_0^1 0.25(2x+y)\,dx = 0.25\Big[x^2 + xy\Big]_0^1 = \boxed{\tfrac{1}{4}(1+y)}, \quad 0 < y < 2$$

Always sanity check that each integrates to 1: $\int_0^1 (x + \frac{1}{2})\,dx = \frac{1}{2} + \frac{1}{2} = 1$ ✓ and $\int_0^2 \frac{1}{4}(1+y)\,dy = \frac{1}{4}(2+2) = 1$ ✓

(c) Find $\mathbb{E}[X]$

Use the marginal, not the joint (either works, the marginal is faster).

$$\mathbb{E}[X] = \int_0^1 x\left(x + \tfrac{1}{2}\right) dx = \int_0^1 \left(x^2 + \tfrac{x}{2}\right) dx = \left[\frac{x^3}{3} + \frac{x^2}{4}\right]_0^1 = \frac{1}{3} + \frac{1}{4} = \boxed{\frac{7}{12}}$$

Remember this number. Part (j) hands it back to you for free.

(d) Are X and Y independent?

Expand: $f_{XY}(x,y) = 0.25(2x+y) = 0.5x + 0.25y$. This is a **sum**, and no sum of that form factors into $g(x)h(y)$. By the **Factorization Theorem (Lemma 4.2.7)**, X and Y are **not independent**.

Alternative phrasing that also earns full credit: $f_X(x)f_Y(y) = (x+\frac{1}{2})\cdot\frac{1}{4}(1+y) \neq 0.25(2x+y)$, so the definition of independence fails.

(e) and (f) The two conditionals

(e) $f_{Y|X}(y|X=x)$**.** Joint over marginal of the *conditioning* variable X:

$$f_{Y|X}(y|x) = \frac{f_{XY}(x,y)}{f_X(x)} = \frac{0.25(2x+y)}{x+0.5}, \qquad \boxed{0<y<2}$$

(f) $f_{X|Y}(x|Y=y)$**.** Now divide by the marginal of Y:

$$f_{X|Y}(x|y) = \frac{f_{XY}(x,y)}{f_Y(y)} = \frac{0.25(2x+y)}{0.25(1+y)} = \frac{2x+y}{1+y}, \qquad \boxed{0<x<1}$$

The support flips, and that is the point

Look at the boxed ranges above. For $f_{Y|X}(y|x)$ the free variable is y, so the range is $0<y<2$. For $f_{X|Y}(x|y)$ the free variable is x, so the range is $0<x<1$. Writing the wrong range here is a guaranteed deduction, and it is easy to do on autopilot. **The range always belongs to the variable on the left of the bar.**

(g) Find $\mathrm{P}(X<0.25 \mid Y=1)$

Conditioning on a specific value means: take the conditional density from (f), plug in $y=1$, then integrate over the x region you want.

$$f_{X|Y}(x|Y=1) = \frac{2x+1}{1+1} = \frac{2x+1}{2} = x+\tfrac{1}{2}, \qquad 0<x<1$$

$$\mathrm{P}(X<0.25 \mid Y=1) = \int_0^{1/4} (x+\tfrac{1}{2})\,dx = \left[\frac{x^2}{2}+\frac{x}{2}\right]_0^{1/4} = \frac{1}{32}+\frac{1}{8} = \boxed{\frac{5}{32} = 0.15625}$$

(h) Find $\mathbb{E}[X \mid Y=y]$

Use the conditional from (f). X **is the random one;** y **is fixed.** So integrate over x, from 0 to 1.

$$\mathbb{E}[X|Y=y] = \int_0^1 x\cdot\frac{2x+y}{1+y}\,dx = \frac{1}{1+y}\int_0^1 (2x^2+xy)\,dx$$

Pull $\frac{1}{1+y}$ out because it has no x in it, which is exactly why identifying the random variable matters.

$$= \frac{1}{1+y}\left[\frac{2x^3}{3}+\frac{x^2y}{2}\right]_0^1 = \boxed{\frac{1}{1+y}\left(\frac{2}{3}+\frac{y}{2}\right)}$$

Note the answer is a **function of** y. That is correct and expected.

(i) Write the integral for the outer expectation

The outer expectation is taken over Y, using the **marginal of** Y from part (b), because after part (h) the surviving variable is y.

$$\mathbb{E}\big[\mathbb{E}[X|Y=y]\big] = \int_0^2 \mathbb{E}[X|Y=y]\, f_Y(y)\, dy = \int_0^2 \frac{1}{1+y}\left(\frac{2}{3}+\frac{y}{2}\right)\cdot\frac{1}{4}(1+y)\, dy$$

That is the complete answer to (i). Stop there, as instructed.

Aside worth noticing: the $(1+y)$ terms cancel, which is why this one happens to be doable. In general it would not be.

(j) Now evaluate it using a result from class

By the **Law of Iterated Expectations (Thm. 4.4.3)**:

$$\mathbb{E}\big[\mathbb{E}[X|Y]\big] = \mathbb{E}[X]$$

And from part (c), $\mathbb{E}[X] = \frac{7}{12}$. So the answer is $\boxed{\frac{7}{12}}$ with no integration at all.

The Law of Iterated Expectations, in words

$$\mathbb{E}[Y] = \mathbb{E}_X\big[\mathbb{E}[Y|X]\big]$$

Average the group averages, weighted by group size, and you recover the overall average.

Dr. Flores's example: let Y be exam grade, $X = 1$ if female and 0 if male. Then

$$\mathbb{E}[Y] = \mathbb{E}[Y|X=0]\mathrm{P}(X=0) + \mathbb{E}[Y|X=1]\mathrm{P}(X=1)$$

Mean grade = (mean for men)(fraction men) + (mean for women)(fraction women). Obviously true, and that is the whole theorem.

Exam signal: whenever you see a nasty integral that turns out to be $\mathbb{E}$ of a conditional expectation, the intended answer is LIE, not calculus. Part (i) and (j) are literally designed as "do it the hard way" and "now do it the smart way."

9 Block 5. Covariance, correlation, and variance of sums

25 minutes

Dr. Flores's notes single this out: *"using formulas for* $\mathrm{Var}(aX+bY)$ *(notes IX, 5.2) and covariance formulas (pg 12). Example: what is* $\mathrm{Corr}(W,U)$*? Use formulas."* This is a mechanical question. You are not being asked to be clever, you are being asked to expand correctly.

9.1 Definitions

Covariance and correlation

$\mathrm{Cov}(X,Y) = \mathbb{E}\big[(X-\mu_X)(Y-\mu_Y)\big]$ "do they tend to be above/below their means together?"

Positive: when X is high, Y tends to be high. Negative: opposites. Zero: no *linear* tendency.

$$\rho_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y}$$ covariance, standardized

Why bother with ρ? Covariance is unit-dependent, so "Cov $= 1000$" tells you nothing without knowing the units. Correlation is unitless and always sits in $[-1,1]$, so "$\rho = 0.95$" immediately means strong linear association. $|\rho| = 1$ exactly when $Y = aX + b$ for some constants.

9.2 The formula sheet you will actually use

Covariance algebra. These are the exam's bread and butter.

(C1) $\mathrm{Cov}(X,X) = \mathrm{Var}(X)$

(C2) $\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$

(C3) $\mathrm{Cov}(X,a) = 0$ for any constant a

(C4) $\mathrm{Cov}(aX+bY,\ cU+dV) = ac\mathrm{Cov}(X,U) + ad\mathrm{Cov}(X,V) + bc\mathrm{Cov}(Y,U) + bd\mathrm{Cov}(Y,V)$

(C5) $\mathrm{Var}(aX+bY) = a^2\mathrm{Var}(X) + b^2\mathrm{Var}(Y) + 2ab\mathrm{Cov}(X,Y)$

(C6) If $X \perp\!\!\!\perp Y$: $\mathrm{Var}(aX+bY) = a^2\mathrm{Var}(X) + b^2\mathrm{Var}(Y)$

(C4) is just FOIL, and that is genuinely how to remember it

Covariance is **bilinear**: it distributes over sums and pulls constants out of each slot separately. So expanding $\mathrm{Cov}(aX+bY,\ cU+dV)$ works exactly like multiplying out $(aX+bY)(cU+dV)$, except every product of two variables becomes a Cov.

And (C5) is just (C4) applied to $\mathrm{Cov}(aX+bY,\ aX+bY)$:

$$\begin{aligned} &a\cdot a\mathrm{Cov}(X,X) + ab\mathrm{Cov}(X,Y) + ba\mathrm{Cov}(Y,X) + b\cdot b\mathrm{Cov}(Y,Y) \\ &\quad = a^2\mathrm{Var}(X) + 2ab\mathrm{Cov}(X,Y) + b^2\mathrm{Var}(Y) \end{aligned}$$

You only need to remember (C4). Everything else falls out of it. That is one formula, not six.

Worked: the lecture example, mechanics drill

Given $\text{Cov}(X,Y) = 4$, $\text{Var}(X) = 2$, $\text{Var}(Y) = 3$. Let $W = 2X + 4$ and $U = 3X + 2Y$.

Find $\text{Cov}(W,U)$**.** Write $W = 2X + 0 \cdot Y + 4$ and apply (C4), remembering the constant 4 contributes nothing by (C3):

$$\text{Cov}(2X+4,\ 3X+2Y) = 2\cdot 3\,\text{Cov}(X,X) + 2\cdot 2\,\text{Cov}(X,Y) = 6\text{Var}(X) + 4\text{Cov}(X,Y)$$

$$= 6(2) + 4(4) = \boxed{28}$$

Find $\text{Var}(U)$**.** Apply (C5) with $a = 3$, $b = 2$:

$$\text{Var}(3X+2Y) = 9\text{Var}(X) + 4\text{Var}(Y) + 2(3)(2)\text{Cov}(X,Y) = 9(2) + 4(3) + 12(4) = 18 + 12 + 48 = \boxed{78}$$

Find $\text{Var}(W)$**.** Just (V2): $\text{Var}(2X+4) = 4\text{Var}(X) = 8$.

A sanity check worth knowing. If you now compute $\text{Corr}(W,U) = 28/\sqrt{8\cdot 78} = 28/\sqrt{624} \approx 1.12$, which is impossible since correlations live in $[-1,1]$. That is because the given numbers are not a legitimate covariance structure: $\rho_{XY} = 4/\sqrt{2\cdot 3} \approx 1.63 > 1$ to begin with. The numbers are a mechanics drill, not a real distribution. **But learning to spot $|\rho| > 1$ as an impossibility is itself an exam-grade insight**, so if you ever produce such a number on a real problem, you have made an arithmetic error.

Worked: PS6 #4, the "find $\text{Corr}(W,U)$" archetype

(X,Y) bivariate normal with parameters $\mu_X, \mu_Y, \sigma_X^2, \sigma_Y^2, \rho$. Let $U = X + Y$ and $W = X - Y$. Find ρ_{UW}.

Step 1. Numerator, using (C4). Note $\text{Cov}(X,Y) = \rho\sigma_X\sigma_Y$.

$$\text{Cov}(U,W) = \text{Cov}(X+Y,\ X-Y) = \text{Var}(X) - \text{Cov}(X,Y) + \text{Cov}(Y,X) - \text{Var}(Y) = \sigma_X^2 - \sigma_Y^2$$

The two cross terms cancel exactly. Nice.

Step 2. The two variances, using (C5).

$$\text{Var}(U) = \sigma_X^2 + \sigma_Y^2 + 2\rho\sigma_X\sigma_Y, \qquad \text{Var}(W) = \sigma_X^2 + \sigma_Y^2 - 2\rho\sigma_X\sigma_Y$$

Step 3. Divide. The denominator is a difference of squares:

$$\text{Var}(U)\text{Var}(W) = \left(\sigma_X^2 + \sigma_Y^2\right)^2 - 4\rho^2\sigma_X^2\sigma_Y^2$$

$$\boxed{\rho_{UW} = \frac{\sigma_X^2 - \sigma_Y^2}{\left[\left(\sigma_X^2 + \sigma_Y^2\right)^2 - 4\rho^2\sigma_X^2\sigma_Y^2\right]^{1/2}}}$$

Read the result. If $\sigma_X^2 = \sigma_Y^2$ then $\rho_{UW} = 0$: the sum and difference of two equally-variable normals are uncorrelated. That is a genuinely useful fact and a good way to check you did it right.

The universal "find the correlation of two linear combinations" procedure

1. Write both new variables in the form $aX + bY$. Constants can be dropped immediately; they never affect covariance or variance.
2. Numerator: expand Cov with (C4), FOIL style. Replace $\text{Cov}(X, X) \to \text{Var}(X)$ and $\text{Cov}(X, Y) \to \rho\sigma_X\sigma_Y$ if you were given ρ instead of a covariance.
3. Denominator: get each variance with (C5), then multiply and take the square root.
4. Divide. Sanity check that the magnitude is at most 1.

10 Session 1 checkpoint

15 minutes

Answer from memory, no notes. If you get 10 of 12, you are on track for the 80%.

1. In $f_{Y|X}(y|x)$, which variable do you integrate over, and what is the answer a function of?
2. Write the change-of-variable formula, including the piece people forget.
3. $X \sim N(2, 9)$, $Y = -3X + 1$. What is the distribution of Y?
4. Give the fastest formula for $\text{Var}(X)$.
5. You are given $f_{XY}(x, y)$ on $0 < x < 1, 0 < y < 3$. Write the integral for $f_Y(y)$, with limits.
6. State the Factorization Theorem and use it on $f_{XY} = 3x^2y^4$.
7. What is $\mathbb{E}\big[\mathbb{E}[Y|X]\big]$, and what is the theorem called?
8. Expand $\text{Var}(2X - 3Y)$.
9. Expand $\text{Cov}(X + 2Y,\ 3X - Y)$.
10. Does $\text{Cov}(X, Y) = 0$ imply independence? When does it?
11. $X \sim N(1, 4)$. To compute $\text{P}(X \leq 3)$, what exactly do you divide by?
12. A cdf answer is missing something. What is almost always missing?

Answers

1. Over y; the answer is a function of x. **2.** $f_Y(y) = f_X(g^{-1}(y)) \left|\frac{\mathrm{d}g^{-1}(y)}{\mathrm{d}y}\right|$ for $y \in \mathcal{Y}$, zero otherwise. The forgotten pieces: the absolute value and the support. **3.** $N(-3(2) + 1,\ 9 \cdot 9) = N(-5, 81)$. **4.** $\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$. **5.** $f_Y(y) = \int_0^1 f_{XY}(x, y)\,\mathrm{d}x$ for $0 < y < 3$. **6.** Independent iff $f_{XY} = g(x)h(y)$; here $(3x^2)(y^4)$ splits, so independent. **7.** $\mathbb{E}[Y]$; Law of Iterated Expectations. **8.** $4\text{Var}(X) + 9\text{Var}(Y) - 12\text{Cov}(X, Y)$. **9.** $3\text{Var}(X) - \text{Cov}(X, Y) + 6\text{Cov}(Y, X) - 2\text{Var}(Y) = 3\text{Var}(X) + 5\text{Cov}(X, Y) - 2\text{Var}(Y)$. **10.** No. It only implies independence when (X, Y) is bivariate normal. **11.** By $\sigma = 2$, not by 4. **12.** The 0 branch and the 1 branch; a cdf must be defined on all of $\mathbb{R}$.

SESSION TWO

Five hours to one hundred

35 min	Block 6	Sets, axioms, and the proof toolkit
40 min	Block 7	Conditional probability, Bayes, independence
35 min	Block 8	The distribution zoo
40 min	Block 9	Moment generating functions
25 min	Block 10	Markov, Chebyshev, Jensen
40 min	Block 11	The bivariate normal and iterated expectations
20 min	Block 12	The strict-grading checklist
45 min	Block 13	The real Sample Midterm, fully solved
60 min	Block 14	Mock midterm and harder bonus problems

11 Block 6. Sets, axioms, and the proof toolkit

35 minutes

This material (Notes I and II, Problem Set 1) is unlikely to be a full question, but it underwrites everything else and it is where Bayes-style questions get their vocabulary. Move briskly.

11.1 The set operations, decoded

Translating set symbols into English

Symbol	English	Probability meaning
$A \cup B$	A or B (or both)	at least one occurs
$A \cap B$	A and B	both occur
A^c	not A	A fails to occur
$A \subset B$	A is inside B	A occurring forces B to occur
$A \cap B = \emptyset$	disjoint, mutually exclusive	they cannot both happen
$A \backslash B$	A but not B	same as $A \cap B^c$

The laws worth having memorized

Commutative, associative, distributive: behave like arithmetic, with $\cup$ acting like $+$ and $\cap$ like $\times$, except the distributive law works *both* ways:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C), \qquad A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

DeMorgan's laws (the ones that actually get used in proofs):

$$(A \cup B)^c = A^c \cap B^c, \qquad (A \cap B)^c = A^c \cup B^c$$

In words: the negation of "or" is "and of the negations". "Not (rain or snow)" equals "no rain and no snow".

Three decomposition identities used constantly in proofs:

$$A = (A \backslash B) \cup (A \cap B), \qquad A \cup B = A \cup (B \cap A^c), \qquad B = (A \cap B) \cup (A^c \cap B)$$

Each one splits a set into *disjoint* pieces, which is exactly what lets you add probabilities.

11.2 Partitions: the concept Bayes runs on

What a partition is

Sets $A_1, A_2, \ldots$ form a **partition** of Ω if two things hold:

1. They are pairwise disjoint: $A_i \cap A_j = \emptyset$ for $i \neq j$. *No overlap.*
2. They cover everything: $\bigcup_i A_i = \Omega$. *No gaps.*

Picture a pie cut into slices. Every point is in exactly one slice.

Examples: {male, female}; $\{D, \neg D\}$ for a disease; {Seattle, Las Vegas, New York} in PS2 #5. Any A and A^c form a partition, always.

Why it matters: a partition lets you break any event into non-overlapping chunks and add. That is the Law of Total Probability, which is the engine inside Bayes' rule.

11.3 Kolmogorov's axioms and their consequences

The three axioms

P is a probability function on $(S, \mathcal{B})$ if:

(a) $\mathrm{P}(A) \geq 0$ for every event A

(b) $\mathrm{P}(S) = 1$

(c) For pairwise *disjoint* $A_1, A_2, \ldots$: $\mathrm{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathrm{P}(A_i)$

That is the entire foundation. Everything below is a consequence.

The eight consequences, and the one-line reason for each

1.	$\mathrm{P}(A^c) = 1 - \mathrm{P}(A)$	A and A^c are disjoint and fill S
2.	$\mathrm{P}(A) \leq 1$	because $\mathrm{P}(A^c) \geq 0$
3.	$\mathrm{P}(\varnothing) = 0$	complement of S
4.	$\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$	subtract the double-counted overlap
5.	$A \subseteq B \Rightarrow \mathrm{P}(A) \leq \mathrm{P}(B)$	$\mathrm{P}(B) - \mathrm{P}(A) = \mathrm{P}(B \cap A^c) \geq 0$
6.	For a partition $C_1, C_2, \ldots$: $\mathrm{P}(A) = \sum_i \mathrm{P}(A \cap C_i)$	chop A into disjoint slices
7.	**Boole**: $\mathrm{P}\left(\bigcup_i A_i\right) \leq \sum_i \mathrm{P}(A_i)$	overlaps only shrink the union
8.	**Bonferroni**: $\mathrm{P}\left(\bigcap_i A_i\right) \geq 1 - \sum_i [1 - \mathrm{P}(A_i)]$	Boole applied to complements

Items 7 and 8 exist to *bound* probabilities when you cannot compute them exactly.

Worked: PS1 #3, the hardest problem in that set

Show $\mathrm{P}(A \cap B) = 1 - \mathrm{P}(A^c) - \mathrm{P}(B^c) + \mathrm{P}(A^c \cap B^c)$.

Strategy: get from $A \cap B$ to complements using DeMorgan. Start with consequence 1:

$$\mathrm{P}(A \cap B) = 1 - \mathrm{P}\big((A \cap B)^c\big)$$

By DeMorgan, $(A \cap B)^c = A^c \cup B^c$. Apply consequence 4 to that union:

$$\mathrm{P}(A^c \cup B^c) = \mathrm{P}(A^c) + \mathrm{P}(B^c) - \mathrm{P}(A^c \cap B^c)$$

Substitute back:

$$\mathrm{P}(A \cap B) = 1 - \Big[\mathrm{P}(A^c) + \mathrm{P}(B^c) - \mathrm{P}(A^c \cap B^c)\Big] = 1 - \mathrm{P}(A^c) - \mathrm{P}(B^c) + \mathrm{P}(A^c \cap B^c) \quad \blacksquare$$

Counting probabilities in the equally-likely case

When S is finite and every outcome is equally likely:

$$\mathrm{P}(A) = \frac{|A|}{|S|} = \frac{\text{number of ways } A \text{ can happen}}{\text{total number of outcomes}}$$

Two dice give $|S| = 36$. PS1 #2 and PS2 #3 are entirely this, plus careful listing.

12 Block 7. Conditional probability, Bayes, and independence

40 minutes

Dr. Flores lists **Bayes' rule** as an alternate question that may appear. It is cheap to prepare and it is Problem Set 2's whole second half.

12.1 Conditional probability

What conditioning actually does

$$\mathrm{P}(A|B) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)}, \qquad \mathrm{P}(B) > 0$$

You are shrinking the universe. Once you know B happened, B becomes the new sample space. The numerator asks "how much of A survives inside B", and the denominator rescales so the new universe has total probability 1.

Coin example: $S = \{HH, HT, TH, TT\}$, $A = \{HH\}$, $B = \{\text{at least one head}\} = \{HH, HT, TH\}$. Then $\mathrm{P}(A|B) = \frac{1/4}{3/4} = \frac{1}{3}$. And indeed, once you restrict to the three outcomes in B, one of them is HH.

Two conditional-probability traps

1. $\mathrm{P}(A|B) + \mathrm{P}(A|B^c) \neq 1$ in general. **What does sum to 1 is** $\mathrm{P}(A|B) + \mathrm{P}(A^c|B)$**.** The thing that must sum to 1 is the piece on the *left* of the bar, not the right. This is exactly the mistake the notes call out in the disease example.
2. $\mathrm{P}(A|B) \neq \mathrm{P}(B|A)$. Confusing them is the base-rate fallacy, and inverting them properly is what Bayes' rule is for.

12.2 The three uses of conditioning

1. The multiplication rule, for joint events

$$\mathrm{P}(A \cap B) = \mathrm{P}(B) \cdot \mathrm{P}(A|B)$$

Chained for many events:

$$\mathrm{P}\left(\bigcap_{i=1}^{n} A_i\right) = \mathrm{P}(A_1)\mathrm{P}(A_2|A_1)\mathrm{P}(A_3|A_1 \cap A_2) \cdots$$

Use case: sampling without replacement. Three red balls from a box with r red and b blue:

$$\mathrm{P}(\text{all red}) = \frac{r}{r+b} \cdot \frac{r-1}{r-1+b} \cdot \frac{r-2}{r-2+b}$$

Each factor updates because a ball was removed. That is the whole trick.

2. Law of Total Probability

For a partition $A_1, \ldots, A_n$ of S and any event B:

$$\mathrm{P}(B) = \sum_{i=1}^{n} \mathrm{P}(B|A_i) \cdot \mathrm{P}(A_i)$$

In words: to get the overall rate, average the group-specific rates weighted by group size. $\mathrm{P}(\text{white}) = \mathrm{P}(\text{white}|\text{female})\mathrm{P}(\text{female}) + \mathrm{P}(\text{white}|\text{male})\mathrm{P}(\text{male})$. **Notice the structural echo:** this is the discrete version of the Law of Iterated Expectations from Block 4. Same idea, one for probabilities, one for means.

3. Bayes' rule, the inversion theorem

$$\boxed{\mathrm{P}(A_i|B) = \frac{\mathrm{P}(B|A_i)\mathrm{P}(A_i)}{\sum_j \mathrm{P}(B|A_j)\mathrm{P}(A_j)}}$$

Without a partition, the simple form:

$$\mathrm{P}(A|B) = \frac{\mathrm{P}(B|A)\mathrm{P}(A)}{\mathrm{P}(B)}$$

The denominator is just the Law of Total Probability. That is the only reason it looks complicated. Structurally Bayes is: numerator is one term, denominator is the sum of all such terms.

Recognition signal: you are handed $\mathrm{P}(B|A)$ and asked for $\mathrm{P}(A|B)$. The conditioning bar flips.

Worked: the diagnostic test, the classic Bayes setup

Let D be disease, $+$ a positive test. Given: sensitivity $\mathrm{P}(+|D) = 0.9$, specificity $\mathrm{P}(-|\neg D) = 0.95$, prevalence $\mathrm{P}(D) = 0.01$. Find $\mathrm{P}(D|+)$.

First, the piece you must derive. You need $\mathrm{P}(+|\neg D)$, and it comes from the *same* conditioning side:

$$\mathrm{P}(+|\neg D) + \mathrm{P}(-|\neg D) = 1 \;\Rightarrow\; \mathrm{P}(+|\neg D) = 1 - 0.95 = 0.05$$

Careful: it is *not* true that $\mathrm{P}(+|\neg D) + \mathrm{P}(+|D) = 1$. The notes flag this explicitly.

Now apply Bayes with the partition $\{D, \neg D\}$:

$$\mathrm{P}(D|+) = \frac{\mathrm{P}(+|D)\mathrm{P}(D)}{\mathrm{P}(+|D)\mathrm{P}(D) + \mathrm{P}(+|\neg D)\mathrm{P}(\neg D)} = \frac{(0.9)(0.01)}{(0.9)(0.01) + (0.05)(0.99)} = \frac{0.009}{0.0585} \approx 0.154$$

Interpret it. A 90% accurate test came back positive and there is still only a 15% chance you have the disease, because the disease is rare. That is the base-rate effect, and it is the reason this example exists.

Worked: PS2 #4, the two-box problem

Box I: 3 red, 7 blue. Box II: 6 red, 4 blue. Pick a box at random, then one ball. **(a)** $\mathrm{P}(\textbf{red})$, by Law of

Total Probability with partition $\{I, II\}$:

$$\mathrm{P}(R) = \mathrm{P}(R|I)\mathrm{P}(I) + \mathrm{P}(R|II)\mathrm{P}(II) = \tfrac{3}{10} \cdot \tfrac{1}{2} + \tfrac{6}{10} \cdot \tfrac{1}{2} = \boxed{\tfrac{9}{20}}$$

(b) $\mathrm{P}(II|\textbf{red})$, by Bayes:

$$\mathrm{P}(II|R) = \frac{\mathrm{P}(R|II)\mathrm{P}(II)}{\mathrm{P}(R)} = \frac{(6/10)(1/2)}{9/20} = \frac{3/10}{9/20} = \boxed{\tfrac{2}{3}}$$

Note the denominator was already computed in part (a). That is the standard structure: part (a) builds the denominator, part (b) does the inversion.

Worked: PS2 #5, Bayes with three branches

Friend visits Seattle, Las Vegas, or New York, each with probability $\frac{1}{3}$. Rain probabilities: $\frac{1}{3}$, $\frac{1}{10}$, $\frac{1}{5}$. She came back wet. Where was she? Let W be "got wet". Joint probabilities:

$$\mathrm{P}(S \cap W) = \tfrac{1}{3} \cdot \tfrac{1}{3} = \tfrac{1}{9}, \quad \mathrm{P}(L \cap W) = \tfrac{1}{3} \cdot \tfrac{1}{10} = \tfrac{1}{30}, \quad \mathrm{P}(N \cap W) = \tfrac{1}{3} \cdot \tfrac{1}{5} = \tfrac{1}{15}$$

$$\mathrm{P}(W) = \tfrac{1}{9} + \tfrac{1}{30} + \tfrac{1}{15} = \tfrac{10+3+6}{90} = \tfrac{19}{90}$$

$$\mathrm{P}(S|W) = \frac{1/9}{19/90} = \boxed{\tfrac{10}{19}}, \quad \mathrm{P}(L|W) = \boxed{\tfrac{3}{19}}, \quad \mathrm{P}(N|W) = \boxed{\tfrac{6}{19}}$$

Check: they sum to 1. That check is free and it catches most errors.

12.3 Independence of events

Pairwise versus mutual independence

Events A and B are independent if $\mathrm{P}(A \cap B) = \mathrm{P}(A)\mathrm{P}(B)$, equivalently $\mathrm{P}(A|B) = \mathrm{P}(A)$: knowing B tells you nothing about A.

For **three or more** events, mutual independence requires the product rule to hold for *every subcollection*:

$$\mathrm{P}(E_1 \cap E_2) = \mathrm{P}(E_1)\mathrm{P}(E_2), \quad \mathrm{P}(E_1 \cap E_3) = \ldots, \quad \mathrm{P}(E_2 \cap E_3) = \ldots,$$
$$\textit{and} \quad \mathrm{P}(E_1 \cap E_2 \cap E_3) = \mathrm{P}(E_1)\mathrm{P}(E_2)\mathrm{P}(E_3)$$

Neither condition implies the other. PS2 #3 is built to prove exactly this:

- Part (a) and (b): three dice events that are *pairwise* independent but *not* mutually independent (the triple intersection is empty, so its probability is 0, not $\frac{1}{8}$).
- Part (c): three events where the *triple* product works but a *pair* fails. So mutual independence fails too.

Exam takeaway: if asked whether events are mutually independent, you must check all of the conditions, and finding one failure is enough to say no.

Useful consequence

If A and B are independent, then so are (A^c, B), (A, B^c), and (A^c, B^c). Independence survives complementation.

13 Block 8. The distribution zoo

35 minutes

You do not need to memorize every pdf. You need to (1) recognize which distribution a story describes, and (2) know the means and variances. Both go on your cheat sheet.

13.1 Discrete distributions

Name	pmf	$\mathbb{E}[X]$	$\mathrm{Var}(X)$	The story
Discrete Uniform on $\{1,\dots,N\}$	$\frac{1}{N}$	$\frac{N+1}{2}$	$\frac{N^2-1}{12}$	Every outcome equally likely. A die.
Bernoulli(p)	$p^x(1-p)^{1-x}$, $x\in\{0,1\}$	p	$p(1-p)$	One trial, success or failure.
Binomial(n,p)	$\binom{n}{x}p^x(1-p)^{n-x}$	np	$np(1-p)$	Number of successes in n independent trials.
Geometric(p)	$p(1-p)^{x-1}$, $x=1,2,\dots$	$\frac{1}{p}$	$\frac{1-p}{p^2}$	Number of trials until the *first* success.
Negative Binomial(r,p)	$\binom{x-1}{r-1}p^r(1-p)^{x-r}$	$\frac{r(1-p)}{p}$	$\frac{r(1-p)}{p^2}$	Number of trials until the *rth* success.
Hypergeometric(N,M,K)	$\frac{\binom{M}{x}\binom{N-M}{K-x}}{\binom{N}{K}}$			Draw K without replacement from N items of which M are "good".
Poisson(λ)	$\frac{e^{-\lambda}\lambda^x}{x!}$	λ	λ	Counts per interval. Mean equals variance.

The family relationships worth knowing

- Binomial$(1,p)$ = Bernoulli(p). A binomial is a sum of n independent Bernoullis.
- Negative Binomial$(1,p)$ = Geometric(p).
- Poisson approximates Binomial when n is large and p is small, with $\lambda = np$.

Worked: PS5 #2, recognizing the binomial story

An airline has 20 seats and takes 23 reservations. 10% of people do not show. What is P(everyone who shows gets a seat)?

Step 1, identify. Each of 23 reservations is an independent trial with $\mathrm{P}(\text{shows}) = 0.9$. So $X =$ number who show $\sim$ Binomial$(23,\ 0.9)$.

Step 2, translate the question. Everyone is accommodated when $X \le 20$.

$$\mathrm{P}(X\le 20) = 1-\mathrm{P}(X=21)-\mathrm{P}(X=22)-\mathrm{P}(X=23)$$

$$= 1-\binom{23}{21}(0.9)^{21}(0.1)^2-\binom{23}{22}(0.9)^{22}(0.1)-\binom{23}{23}(0.9)^{23}$$

$$= 1-253(0.9)^{21}(0.01)-23(0.9)^{22}(0.1)-(0.9)^{23} = \boxed{0.4080}$$

The exam-relevant skill is going the complement route. Computing $P(X \leq 20)$ directly would be 21 terms; the complement is 3.

Worked: PS5 #3, a Poisson parameter puzzle

$X \sim \text{Poisson}(\lambda)$ with $P(X = 1) = P(X = 2)$. Find $P(X = 3)$. Set the two pmf values equal and solve for λ:

$$\lambda e^{-\lambda} = \frac{\lambda^2 e^{-\lambda}}{2!} \Rightarrow 1 = \frac{\lambda}{2} \Rightarrow \lambda = 2$$

$$P(X = 3) = \frac{e^{-2}2^3}{3!} = \frac{8e^{-2}}{6} = \boxed{0.1804}$$

13.2 Continuous distributions

Name	pdf	$\mathbb{E}[X]$	$\text{Var}(X)$	Notes
Uniform(a, b)	$\frac{1}{b-a}$ on (a, b)	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	Flat.
Gamma(α, β)	$\frac{x^{\alpha-1}e^{-x/\beta}}{\Gamma(\alpha)\beta^\alpha}$, $x > 0$	$\alpha\beta$	$\alpha\beta^2$	Very flexible shape.
Exponential(λ)	$\frac{1}{\lambda}e^{-x/\lambda}$, $x > 0$	λ	λ^2	Waiting time. See warning below.
Chi-square(p)	$\frac{x^{p/2-1}e^{-x/2}}{\Gamma(p/2)2^{p/2}}$, $x > 0$	p	$2p$	$=$ Gamma$(p/2, 2)$.
Beta(α, β)	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	On $(0, 1)$. Good for proportions.
Normal(μ, σ^2)	$\frac{1}{\sqrt{2\pi\sigma^2}}e^{-(x-\mu)^2/(2\sigma^2)}$	μ	σ^2	The main event.
t with r df	(see notes)	0 if $r > 1$	$\frac{r}{r-2}$ if $r > 2$	Fat tails. No mgf. $r = 1$ is Cauchy, no mean.
$F(r_1, r_2)$	(see book)	exists if $r_2 > 2$	exists if $r_2 > 4$	$F(1, r_2) = t^2_{(r_2)}$.

The exponential parameterization warning. Read this carefully.

Your Notes VIII §3.4 writes the exponential pdf as $\lambda e^{-\lambda x}$ but then states $\mathbb{E}[X] = \lambda$ and $\text{Var}(X) = \lambda^2$. **Those two statements are inconsistent with each other.**

The convention used *consistently* everywhere else in your course (Notes VII, the mgf derivation, and the identity Gamma$(1, \beta)$ =Exponential(β) which needs $\mathbb{E} = \alpha\beta = \beta$) is the **scale** form:

$$f_X(x) = \frac{1}{\lambda}e^{-x/\lambda}, \qquad \mathbb{E}[X] = \lambda, \qquad \text{Var}(X) = \lambda^2, \qquad M_X(t) = \frac{1}{1-\lambda t} \text{ for } t < \tfrac{1}{\lambda}$$

What to do on the exam: use the professor's convention, $\mathbb{E}[X] = \lambda$. If a problem hands you a density of the form ce^{-cx}, just say "exponential with mean $1/c$" and you are unambiguous and correct either way. Naming the mean is safer than naming the parameter.

The Gamma function, just enough

$$\Gamma(\alpha) = \int_0^\infty u^{\alpha-1} e^{-u}\, \mathrm{d}u$$

- $\Gamma(1) = 1$
- $\Gamma(1/2) = \sqrt{\pi}$
- $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$
- $\Gamma(n) = (n-1)!$ for integer n

Intuition: a continuous extension of the factorial whose job is to make the density integrate to 1.

Key distributional identities

- $\text{Gamma}(1, \beta) = \text{Exponential}(\beta)$
- $\text{Gamma}(p/2,\, 2) = \chi^2_{(p)}$ PS5 #4(a) asks you to prove this with mgfs
- $\text{Beta}(1, 1) = \text{Uniform}(0, 1)$
- $Z \sim N(0,1) \Rightarrow Z^2 \sim \chi^2_{(1)}$ PS5 #4(b)
- Independent chi-squares add, and so do their degrees of freedom: $\sum_i X_i \sim \chi^2_{(\sum_i p_i)}$ PS6 #5
- $X \sim N(\mu, \sigma^2),\ Y = aX + b \Rightarrow Y \sim N(a\mu + b, a^2\sigma^2)$
- $\log X \sim N(\mu, \sigma^2)$ means X is log-normal. Good for income.

14 Block 9. Moment generating functions

40 minutes

What an mgf is and why it exists

$$M_X(t) = \mathbb{E}\left[e^{tX}\right] = \begin{cases} \sum_x e^{tx} f_X(x) & \text{discrete} \\ \int_x e^{tx} f_X(x)\,dx & \text{continuous} \end{cases}$$

t is a dummy variable with no meaning of its own. It is a mathematical handle. The function M_X encodes the entire distribution, and you extract information by differentiating at $t = 0$.

Why it works, in one line: $e^{tX} = 1 + tX + \frac{t^2X^2}{2} + \ldots$, so $\mathbb{E}[e^{tX}]$ has all the moments $\mathbb{E}[X], \mathbb{E}[X^2], \ldots$ baked into its coefficients. Differentiating peels them off one at a time.

14.1 The three uses

Use 1: generate moments (Theorem 2.3.7)

$$\mathbb{E}[X^n] = \left.\frac{d^n M_X(t)}{dt^n}\right|_{t=0}$$

Differentiate n times, then set $t = 0$. First derivative at 0 gives the mean; second gives $\mathbb{E}[X^2]$; then use Var $= \mathbb{E}[X^2] - (\mathbb{E}[X])^2$.

Use 2: identify a distribution (Theorem 2.3.11)

If $M_X(t) = M_Y(t)$ for all t in an interval around 0, then $X \sim Y$: **they have the same distribution**.

This is the workhorse for every "show that X follows a ___ distribution" question. The recipe is always the same:

1. Compute M of the thing you have.
2. Show it equals the known M of the target distribution.
3. **Cite Theorem 2.3.11 by name** and conclude.

Step 3 is worth real credit. Write the sentence.

Use 3: sums of independent variables (Theorem 4.2.12)

If $X \perp\!\!\!\perp Y$ and $Z = X + Y$, then

$$M_Z(t) = M_X(t) \cdot M_Y(t)$$

For n independent variables, multiply all n mgfs. **This turns convolution, which is horrible, into multiplication, which is easy.** It is how you prove that sums of chi-squares are chi-square and sums of normals are normal.

14.2 The mgfs you should have on the cheat sheet

Distribution	$M_X(t)$	Valid for
Bernoulli(p)	$(1-p)+pe^t$	all t
Binomial(n,p)	$\left[(1-p)+pe^t\right]^n$	all t
Poisson(λ)	$e^{\lambda(e^t-1)}$	all t
Exponential(λ), prof's convention	$\dfrac{1}{1-\lambda t}$	$t<1/\lambda$
Gamma(α,β)	$\left(\dfrac{1}{1-\beta t}\right)^{\alpha}$	$t<1/\beta$
$\chi^2_{(p)}$	$\left(\dfrac{1}{1-2t}\right)^{p/2}$	$t<1/2$
Normal(μ,σ^2)	$\exp\left\{\mu t+\dfrac{\sigma^2t^2}{2}\right\}$	all t
Uniform(a,b)	$\dfrac{e^{tb}-e^{ta}}{t(b-a)}$	$t\neq 0$

Uniform is the cautionary tale, PS4 #4

The uniform mgf is **undefined at** $t=0$ (you would divide by zero). Since getting moments requires evaluating derivatives *at* $t=0$, **you cannot use the mgf to get the moments of a uniform**. Compute $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$ directly by integration instead. If an exam question asks "can we use the mgf here?", this is the expected answer.

Worked: normal mgf to mean and variance, PS4 #5(b)

Given $M_X(t)=\exp\left\{\mu t+\frac{\sigma^2t^2}{2}\right\}$.

$$M_X'(t)=\exp\left\{\mu t+\tfrac{\sigma^2t^2}{2}\right\}\cdot(\mu+\sigma^2t)\ \Rightarrow\ \mathbb{E}[X]=M_X'(0)=1\cdot\mu=\mu$$

$$M_X''(t)=\exp\{\cdot\}(\mu+\sigma^2t)^2+\exp\{\cdot\}\sigma^2\ \Rightarrow\ \mathbb{E}[X^2]=M_X''(0)=\mu^2+\sigma^2$$

$$\mathrm{Var}(X)=\mathbb{E}[X^2]-(\mathbb{E}[X])^2=\mu^2+\sigma^2-\mu^2=\boxed{\sigma^2}$$

So the parameters really are the mean and the variance, which is why the notation $N(\mu,\sigma^2)$ is written that way.

Worked: PS5 #4(a), Gamma to Chi-square by mgf matching

Show that Gamma$\left(\frac{p}{2},2\right)=\chi^2_{(p)}$. Take the two mgfs as given. If $X\sim$Gamma(α,β) then $M_X(t)=\left(\frac{1}{1-\beta t}\right)^{\alpha}$. Substitute $\alpha=\frac{p}{2}$ and $\beta=2$:

$$M_X(t)=\left(\frac{1}{1-2t}\right)^{p/2}$$

which is exactly the mgf of a $\chi^2_{(p)}$. By **Theorem 2.3.11**, the two random variables have the same distribution. ■ *Notice how short that is.* Once you have the mgf table, these questions cost thirty seconds.

Worked: PS6 #5, sums of independent chi-squares

Let $X_1, \ldots, X_n$ be independent with $X_i \sim \chi^2_{(p_i)}$, and $Y = \sum_{i=1}^n X_i$. By Theorem 4.2.12, mgfs of independent variables multiply:

$$M_Y(t) = \prod_{i=1}^n M_{X_i}(t) = \prod_{i=1}^n \left(\frac{1}{1-2t}\right)^{p_i/2} = \left(\frac{1}{1-2t}\right)^{\sum_i p_i/2}$$

That is the mgf of a χ^2 with $\sum_i p_i$ degrees of freedom. By Theorem 2.3.11, $Y \sim \chi^2_{(\sum_i p_i)}$. ■ **The pattern to internalize:** multiplying mgfs added the exponents, and the exponents were the degrees of freedom. Degrees of freedom add because exponents add.

Worked: PS5 #4(b), the standard-normal-squared trick

Show $Y = Z^2 \sim \chi^2_{(1)}$ when $Z \sim N(0,1)$.

$$M_Y(t) = \mathbb{E}\left[e^{tZ^2}\right] = \int_{-\infty}^{\infty} e^{tz^2} \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2(1-2t)}{2}}\, dz$$

Now the trick that appears all over this course: **make the integrand into a density so it integrates to 1**. The exponent matches a $N\left(0, \frac{1}{1-2t}\right)$ density, which needs $\frac{1}{\sqrt{2\pi}}\sqrt{1-2t}$ out front. Multiply and divide:

$$= \frac{1}{(1-2t)^{1/2}} \underbrace{\int_{-\infty}^{\infty} \frac{\sqrt{1-2t}}{\sqrt{2\pi}} e^{-\frac{z^2}{2/(1-2t)}}\, dz}_{=1, \text{ it is a density}} = \frac{1}{(1-2t)^{1/2}}$$

which is the $\chi^2_{(1)}$ mgf. By Theorem 2.3.11, $Z^2 \sim \chi^2_{(1)}$. ■

The "make it a density" trick, worth naming

Twice above and once in PS4 #5(a), the same move appears: you are stuck with a hard integral, so you **manipulate the integrand until it is a recognizable pdf**, then declare the integral equals 1 and keep whatever constant you pulled outside. Dr. Flores calls this out explicitly as "a very common trick used in statistics." If an integral looks impossible, ask: *can I make this into a density?*

15 Block 10. Markov, Chebyshev, Jensen

25 minutes

Three inequalities, three distinct jobs. Know which one to reach for.

Markov: bound a tail using only the mean

For a **non-negative** random variable X with finite mean, and any $c > 0$:

$$P(X \geq c) \leq \frac{\mathbb{E}[X]}{c}$$

Requirement: $X \geq 0$**.** This is not optional; the inequality is false without it. *Intuition*: if the average is 2, you cannot have a large chunk of probability sitting out at 100, because that alone would drag the average above 2.

Chebyshev: bound the probability of being far from the mean

For **any** random variable Y with mean μ and variance σ^2, and any $k > 0$:

$$P(|Y - \mu| \geq k) \leq \frac{\sigma^2}{k^2}$$

No distributional assumptions at all. That is what makes it powerful and also why the bound is usually loose. **How it is derived from Markov**: set $X = (Y - \mu)^2$, which is non-negative, and $c = k^2$. Then $\mathbb{E}[X] = \text{Var}(Y)$ and Markov gives exactly Chebyshev. Knowing this derivation is worth a point if asked.

Worked: PS5 #5, where Chebyshev is exactly tight

$P(X = -1) = \frac{1}{8}$, $P(X = 0) = \frac{6}{8}$, $P(X = 1) = \frac{1}{8}$. Find the Chebyshev bound on $P(|X - \mu| \geq 2\sigma)$ and compare to the truth. **Moments first.**

$$\mu = \mathbb{E}[X] = (-1)\tfrac{1}{8} + 0 \cdot \tfrac{6}{8} + 1 \cdot \tfrac{1}{8} = 0$$

$$\mathbb{E}[X^2] = 1 \cdot \tfrac{1}{8} + 0 + 1 \cdot \tfrac{1}{8} = \tfrac{1}{4} \;\Rightarrow\; \sigma^2 = \tfrac{1}{4} - 0 = \tfrac{1}{4} \;\Rightarrow\; \sigma = \tfrac{1}{2}$$

The bound. Chebyshev with $k = 2\sigma$:

$$P(|X - \mu| \geq 2\sigma) \leq \frac{\sigma^2}{(2\sigma)^2} = \frac{1}{4}$$

The truth. $2\sigma = 1$, so we want $P(|X| \geq 1) = P(X = -1) + P(X = 1) = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$. The bound and the actual probability are **both** $\frac{1}{4}$. Chebyshev is exactly attained here, which is unusual and is the point of the exercise. *Useful general fact*: with $k = m\sigma$, Chebyshev reads $P(|X - \mu| \geq m\sigma) \leq \frac{1}{m^2}$.

Jensen: compare $\mathbb{E}[g(X)]$ to $g(\mathbb{E}[X])$

$$g \text{ convex } \Rightarrow \mathbb{E}[g(X)] \geq g\big(\mathbb{E}[X]\big)$$

$$g \text{ concave } \Rightarrow \mathbb{E}[g(X)] \leq g\big(\mathbb{E}[X]\big)$$

Strict if g is strictly convex or concave. **Two applications you should be able to state instantly:**

- $g(x) = x^2$ is strictly convex, so $\mathbb{E}[X^2] > \big(\mathbb{E}[X]\big)^2$. *Which is exactly the statement* $\mathrm{Var}(X) > 0$.
- $g(x) = \ln x$ is strictly concave, so $\mathbb{E}[\ln X] < \ln\big(\mathbb{E}[X]\big)$.

How to remember which direction: convex means bowl-shaped ($\cup$), and averaging inputs of a bowl-shaped function undershoots, so the expectation of the function is the bigger one. Convex gives $\geq$.

Which inequality do I use?

- "Bound $\mathrm{P}(X \geq c)$" with $X \geq 0$ and only the mean known $\Rightarrow$ **Markov**.
- "Bound the probability that X is at least k away from its mean" $\Rightarrow$ **Chebyshev**.
- "Is $\mathbb{E}[g(X)]$ bigger or smaller than $g(\mathbb{E}[X])$?" $\Rightarrow$ **Jensen**.

16 Block 11. The bivariate normal and iterated expectations

40 minutes

16.1 The bivariate normal: five parameters, four facts

The setup

$(X, Y) \sim N(\mu_X, \mu_Y, \sigma_X^2, \sigma_Y^2, \rho)$. **Five parameters**: two centers, two spreads, one correlation. You will essentially never need to write the joint pdf out. You need the four facts below.

The four facts

Fact 1. The marginals are normal.

$$X \sim N(\mu_X, \sigma_X^2), \qquad Y \sim N(\mu_Y, \sigma_Y^2)$$

The converse is false: two normal marginals do not guarantee a bivariate normal joint.

Fact 2. Zero correlation equals independence, in both directions.

$$\rho = 0 \iff X \perp\!\!\!\perp Y$$

This is special to the bivariate normal. In general $\rho = 0$ does not give independence, as Block 4 stressed. Here it does, and PS6 #6(a) asks you to prove it (set $\rho = 0$ in the joint pdf and watch it factor into two univariate normal densities, then cite the Factorization Theorem).

Fact 3. The conditional distribution is normal, with a formula.

$$Y \mid X \sim N\left(\underbrace{\mu_Y + \rho \frac{\sigma_Y}{\sigma_X}(X - \mu_X)}_{\mathbb{E}[Y|X]}, \underbrace{\sigma_Y^2(1-\rho^2)}_{\mathrm{Var}(Y|X)} \right)$$

Fact 4. Linear combinations are normal.

$$aX + bY \sim N\left(a\mu_X + b\mu_Y,\ a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\,\rho\sigma_X\sigma_Y\right)$$

The mean and variance come straight from Blocks 2 and 5; the only new content is that the shape stays normal. That is PS6 #6(b).

Why Fact 3 is the most important line in the course

Rearrange the conditional mean:

$$\mathbb{E}[Y|X] = \mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(X - \mu_X) = \underbrace{\left(\mu_Y - \mu_X\rho\frac{\sigma_Y}{\sigma_X}\right)}_{a} + \underbrace{\rho\frac{\sigma_Y}{\sigma_X}}_{b} X$$

That is a straight line: $\mathbb{E}[Y|X] = a + bX$**.** And the slope is

$$b = \rho\frac{\sigma_Y}{\sigma_X} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X\sigma_Y}\cdot\frac{\sigma_Y}{\sigma_X} = \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(X)}$$

which is exactly the OLS slope coefficient. This is why linear regression is the natural thing to do, and it is the bridge from this course into GSE 520. Dr. Flores calls it out in the notes with "That is the formula of β in OLS!" Expect it to matter conceptually.

Also notice $\mathrm{Var}(Y|X) = \sigma_Y^2(1-\rho^2)$ does *not* depend on X, and it shrinks as $|\rho|$ grows. Knowing X reduces your uncertainty about Y, and the stronger the correlation, the more it reduces it. At $|\rho| = 1$ the conditional variance is zero: Y is perfectly determined by X.

16.2 Conditional expectation as a tool

The Law of Iterated Expectations, three forms

$$\textbf{Basic:} \quad \mathbb{E}[Y] = \mathbb{E}_X\big[\mathbb{E}[Y|X]\big]$$

$$\textbf{With a function:} \quad \mathbb{E}[r(Y)] = \mathbb{E}_X\big[\mathbb{E}[r(Y)|X]\big]$$

$$\textbf{Nested (Sample Midterm \#6):} \quad \mathbb{E}[Y|X] = \mathbb{E}_{Z|X}\big[\mathbb{E}[Y|X,Z] \;\big|\; X\big]$$

The nested version says: conditioning on more information and then averaging that extra information away gets you back to where you started. **You can always add a variable and then integrate it out.**

Getting the outer expectation right

When you compute $\mathbb{E}\big[\mathbb{E}[Y|X]\big]$, the outer expectation is taken over X, using the marginal $f_X(x)$, because after the inner step the only surviving random variable is X. PS6 #2(i) is testing exactly this, and it is why the answer used $f_Y(y)$ there: the inner step had conditioned on Y, so Y was what survived. **Rule: the outer expectation uses the marginal of whatever was on the right of the bar.**

17 Block 12. The strict-grading checklist

20 minutes

You said you want 100% under strict grading. Content knowledge gets you to roughly 90. This page is the other 10. These are the deductions that happen to people who actually knew the material.

The twelve most expensive small mistakes

1. **Omitting the support.** Every pdf, pmf, and conditional density answer must end with "for $y \in (\dots)$, and zero otherwise." No exceptions.
2. **Incomplete cdf.** A cdf needs the 0 branch and the 1 branch. It is defined on all of $\mathbb{R}$.
3. **Dropping the absolute value** in the change-of-variable formula. A negative density is an automatic flag.
4. **Wrong variable's range on a conditional.** $f_{Y|X}(y|x)$ carries y's range; $f_{X|Y}(x|y)$ carries x's.
5. **Standardizing by σ^2 instead of σ.** Write $\sigma = \sqrt{\sigma^2}$ in the margin the moment you see $N(\mu, \sigma^2)$.
6. **Integrating the wrong variable for a marginal.** To get f_X, integrate out y. If y survives, you did it backwards.
7. **Saying "uncorrelated therefore independent"** without the bivariate normal assumption. Dr. Flores really dislikes this one.
8. **Not naming the theorem.** Write "by the Factorization Theorem (Lemma 4.2.7)", "by Theorem 2.3.11", "by the Law of Iterated Expectations". Free credit.
9. **Computing when a theorem was intended.** If the problem says "use a result from class", a page of integration is the wrong answer even if the number is right.
10. **Forgetting that discrete probabilities care about $\leq$ versus $<$.** $P(X \geq 2) = 1 - P(X \leq 1)$, not $1 - P(X \leq 2)$. For continuous variables it does not matter; for discrete it absolutely does.
11. **Not checking that a density integrates to 1.** It takes fifteen seconds and it catches most algebra errors.
12. **Writing a bare number with no setup.** His instruction is literally "show as much work and logic as possible". Setup earns more than the answer.

Sanity checks that cost seconds and save points

- Does the density integrate to 1 over its support?
- Is $|\rho| \leq 1$? If you computed 1.12, you made an error.
- Is the variance non-negative?
- Are the probabilities in $[0, 1]$?
- Do the posterior probabilities in a Bayes problem sum to 1?
- Does $\mathbb{E}[\mathbb{E}[Y|X]]$ agree with your directly computed $\mathbb{E}[Y]$? *This one double-checks two answers at once and is the single best check on a joint-distribution question.*
- Does the conditional density integrate to 1 over the free variable?

Exam-day time allocation, for a 10-question paper

- **First pass (40% of time)**: do every question you recognize instantly. The $N(\mu, \sigma^2)$ transformation, the Bernoulli variance, the factorization question. These are one to three lines each.

- **Second pass (45%)**: the joint-distribution multi-part question. Budget real time; it is worth four to five points.
- **Third pass (15%)**: whatever is left, plus writing setups for anything unfinished. **An unfinished question with a correct labeled setup is worth far more than a blank.**

18 Block 13. The real Sample Midterm, fully solved

45 minutes

These are Dr. Flores's actual "Sample Questions from Old Midterms". Try each one before reading the solution. His own labels (Easy, Medium, Hard) are preserved, and remember he said he does *not* plan to include Hard questions.

SM #1 (Easy). $X \sim N(1,4)$ and $Y = 2X - 1$. What is the distribution of Y?

The question note says you do not need to derive it, only state it.

$$\mathbb{E}[Y] = 2\mathbb{E}[X] - 1 = 2(1) - 1 = 1 \qquad \text{Var}(Y) = 2^2\text{Var}(X) = 4(4) = 16$$

$$\boxed{Y \sim N(1,16)}$$

Both parameters changed even though the mean happened to land back on 1. Do not pattern-match; compute both.

SM #2 (Easy). Are X_1, X_2, X_3 independent, given the joint pdf below?

$$f_{X_1X_2X_3}(x_1,x_2,x_3) = \exp\{-(x_1+x_2+x_3)\}, \qquad 0 < x_i < \infty$$

Yes, they are independent. Justification, which is the actual content of the question:

$$\exp\{-(x_1+x_2+x_3)\} = e^{-x_1} \cdot e^{-x_2} \cdot e^{-x_3}$$

The joint density factors into a product of functions each depending on only one variable, and the support is the rectangle $(0,\infty)^3$ so no variable's range depends on another. By the **Factorization Theorem (Lemma 4.2.7)**, extended to n variables, X_1, X_2, X_3 are (mutually) independent.

Bonus observation worth writing: each factor e^{-x_i} is an exponential density with mean 1, so each $X_i \sim$ Exponential with $\mathbb{E}[X_i] = 1$.

Why the exponent trick matters: a sum *in an exponent* becomes a product, so $e^{-(a+b)}$ factors even though $-(a+b)$ does not. Compare PS6 #2 where $0.25(2x+y)$ is a bare sum and therefore does *not* factor. Recognizing the difference is the whole skill.

SM #3 (Easy). Show that a Bernoulli random variable has $\text{Var}(X) = p(1-p)$.

Given $f(x) = p^x(1-p)^{1-x}$ for $x \in \{0,1\}$, so $\text{P}(X=1) = p$ and $\text{P}(X=0) = 1-p$. **Step 1, the mean.**

$$\mathbb{E}[X] = \sum_{x=0}^{1} x\,f(x) = 0 \cdot (1-p) + 1 \cdot p = p$$

Step 2, the second moment. Because $0^2 = 0$ and $1^2 = 1$, squaring changes nothing:

$$\mathbb{E}[X^2] = \sum_{x=0}^{1} x^2 f(x) = 0^2(1-p) + 1^2(p) = p$$

Step 3, combine with the shortcut formula.

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2 = p - p^2 = \boxed{p(1-p)} \quad \blacksquare$$

This is the template for every "compute $\mathbb{E}$ and Var" question: mean, second moment, subtract. Write those three lines and you have the point.

SM #4 (Medium). Derive the density of $Y = X^\beta$ when $f(x) = \alpha\beta x^{\beta-1}e^{-\alpha x^\beta}$, $x \geq 0, \alpha, \beta > 0$.

This is a Weibull density, and it is the transformation question Dr. Flores's notes point to. Run the four-step recipe.

Step 1. Support. $x \in [0, \infty)$ and $y = x^\beta$ with $\beta > 0$, so $\mathcal{Y} = [0, \infty)$.

Step 2. Monotone? $\dfrac{dy}{dx} = \beta x^{\beta-1} > 0$ for $x > 0$. Strictly increasing on the support. ✓

Step 3. Invert and differentiate.

$$y = x^\beta \;\Rightarrow\; x = g^{-1}(y) = y^{1/\beta}, \qquad \frac{dg^{-1}(y)}{dy} = \frac{1}{\beta}y^{\frac{1}{\beta}-1}$$

Step 4. Apply the change of variable formula. First evaluate f_X at $y^{1/\beta}$:

$$f_X\big(y^{1/\beta}\big) = \alpha\beta\left(y^{1/\beta}\right)^{\beta-1} e^{-\alpha\left(y^{1/\beta}\right)^\beta} = \alpha\beta\, y^{\frac{\beta-1}{\beta}} e^{-\alpha y}$$

Note $\left(y^{1/\beta}\right)^\beta = y$, which is what makes the exponent collapse. Now multiply by the Jacobian:

$$f_Y(y) = \alpha\beta\, y^{\frac{\beta-1}{\beta}} e^{-\alpha y} \cdot \frac{1}{\beta}y^{\frac{1}{\beta}-1} = \alpha\, y^{\left(1-\frac{1}{\beta}\right)+\left(\frac{1}{\beta}-1\right)} e^{-\alpha y} = \alpha\, y^0 e^{-\alpha y}$$

$$\boxed{f_Y(y) = \alpha e^{-\alpha y}, \qquad y > 0, \qquad 0 \text{ otherwise}}$$

Identify it: an exponential density with $\mathbb{E}[Y] = 1/\alpha$. All the β machinery cancelled exactly, which is the elegant punchline. The exponents summed to zero because $\left(1 - \frac{1}{\beta}\right) + \left(\frac{1}{\beta} - 1\right) = 0$.

To be safe about the course's exponential notation, write your conclusion as "Y follows an exponential distribution with mean $1/\alpha$", which is unambiguous under either parameterization.

SM #5 (Medium). $f(x,y) = x + y$ **on** $x \in (0,1), y \in (0,1)$**. Four parts.**

This is the same machine as PS6 #2. Dr. Flores's notes flag both as standard exam questions. Same steps, different numbers.

Preliminary: verify it is a density and get both marginals.

$$\int_0^1\int_0^1 (x+y)\,dx\,dy = \int_0^1 \left[\tfrac{x^2}{2} + xy\right]_0^1 dy = \int_0^1 \left(\tfrac{1}{2} + y\right) dy = \tfrac{1}{2} + \tfrac{1}{2} = 1 \quad \checkmark$$

$$f_X(x) = \int_0^1 (x+y)\,dy = \left[xy + \tfrac{y^2}{2}\right]_0^1 = x + \tfrac{1}{2}, \qquad 0 < x < 1$$

$$f_Y(y) = \int_0^1 (x+y)\,dx = \left[\tfrac{x^2}{2} + xy\right]_0^1 = y + \tfrac{1}{2}, \qquad 0 < y < 1$$

The problem is symmetric in x and y, so the marginals match. Notice that and save yourself an integral.

(a) Conditional density of Y given X.

$$f_{Y|X}(y|x) = \frac{f_{XY}(x,y)}{f_X(x)} = \boxed{\frac{x+y}{x+\frac{1}{2}}}, \qquad 0 < y < 1, \ \ 0 \text{ otherwise}$$

(b) Verify it is a valid density. Two conditions:

- **Non-negative**: for $x, y \in (0,1)$ both $x + y > 0$ and $x + \frac{1}{2} > 0$, so the ratio is positive. ✓
- **Integrates to 1**, over y because y is the random one:

$$\int_0^1 \frac{x+y}{x+\frac{1}{2}}\,dy = \frac{1}{x+\frac{1}{2}}\int_0^1 (x+y)\,dy = \frac{1}{x+\frac{1}{2}}\left[xy + \tfrac{y^2}{2}\right]_0^1 = \frac{x+\frac{1}{2}}{x+\frac{1}{2}} = 1 \quad \checkmark$$

Note it integrates to 1 for every value of x, which is the point of the rescaling by the marginal.

(c) Conditional expectation of Y given X. Integrate over y; treat x as a constant and pull it out.

$$\mathbb{E}[Y|X=x] = \int_0^1 y \cdot \frac{x+y}{x+\frac{1}{2}}\,dy = \frac{1}{x+\frac{1}{2}}\int_0^1 \left(xy + y^2\right) dy = \frac{1}{x+\frac{1}{2}}\left[\frac{xy^2}{2} + \frac{y^3}{3}\right]_0^1$$

$$= \frac{\frac{x}{2} + \frac{1}{3}}{x+\frac{1}{2}} = \boxed{\frac{3x+2}{6x+3}}$$

(The second form comes from multiplying top and bottom by 6.) The answer is a function of x, as it must be.

Free check via LIE. $\mathbb{E}[Y]$ should equal $\mathbb{E}\big[\mathbb{E}[Y|X]\big]$. Since $6x + 3 = 6(x + \frac{1}{2})$:

$$\int_0^1 \frac{3x+2}{6(x+\frac{1}{2})}\left(x+\tfrac{1}{2}\right) dx = \frac{1}{6}\int_0^1 (3x+2)\,dx = \frac{1}{6}\left(\tfrac{3}{2} + 2\right) = \frac{7}{12}$$

And directly, $\mathbb{E}[Y] = \int_0^1 y\left(y + \frac{1}{2}\right) dy = \frac{1}{3} + \frac{1}{4} = \frac{7}{12}$. ✓ *Same $\frac{7}{12}$ as PS6 #2(c). Nice coincidence, useful mnemonic.*

(d) The correlation coefficient ρ_{XY}. Assemble five ingredients.

$$\mathbb{E}[X] = \mathbb{E}[Y] = \int_0^1 x\left(x+\tfrac{1}{2}\right) dx = \left[\frac{x^3}{3} + \frac{x^2}{4}\right]_0^1 = \frac{1}{3} + \frac{1}{4} = \frac{7}{12}$$

$$\mathbb{E}[X^2] = \int_0^1 x^2\left(x+\tfrac{1}{2}\right)\,dx = \left[\frac{x^4}{4}+\frac{x^3}{6}\right]_0^1 = \frac{1}{4}+\frac{1}{6} = \frac{5}{12}$$

$$\mathrm{Var}(X) = \mathrm{Var}(Y) = \frac{5}{12} - \left(\frac{7}{12}\right)^2 = \frac{60}{144} - \frac{49}{144} = \frac{11}{144}$$

$$\mathbb{E}[XY] = \int_0^1\int_0^1 xy(x+y)\,dx\,dy = \int_0^1\int_0^1 \left(x^2y+xy^2\right)\,dx\,dy = \int_0^1 \left(\frac{y}{3}+\frac{y^2}{2}\right)\,dy = \frac{1}{6}+\frac{1}{6} = \frac{1}{3}$$

$$\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = \frac{1}{3} - \frac{49}{144} = \frac{48}{144} - \frac{49}{144} = -\frac{1}{144}$$

$$\rho_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X\sigma_Y} = \frac{-1/144}{\sqrt{\frac{11}{144}}\sqrt{\frac{11}{144}}} = \frac{-1/144}{11/144} = \boxed{-\frac{1}{11} \approx -0.0909}$$

Sanity check: $|\rho| \le 1$. ✓ It is slightly negative, which makes sense: the density must integrate to 1 over a fixed square, so large x mildly crowds out large y.

SM #6 (Hard, and he said he will not include one). The nested iterated expectation

This is the nested Law of Iterated Expectations from Notes IX §6.1. Continuous case; swap integrals for sums if discrete. Start from the right-hand side. The outer expectation is over Z given $X = x$, so it uses $f_{Z|X}(z|x)$:

$$\mathbb{E}\Big\{\mathbb{E}[Y|X,Z] \;\Big|\; X=x\Big\} = \int_z \underbrace{\left[\int_y y\, f_{Y|X,Z}(y|x,z)\,dy\right]}_{\mathbb{E}[Y|X=x,Z=z]} f_{Z|X}(z|x)\,dz$$

Swap the order of integration and group the two densities:

$$= \int_y y\left[\int_z f_{Y|X,Z}(y|x,z)\, f_{Z|X}(z|x)\,dz\right]\,dy$$

Now simplify the bracket using the definition of conditional density twice:

$$f_{Y|X,Z}(y|x,z)\, f_{Z|X}(z|x) = \frac{f_{XYZ}(x,y,z)}{f_{XZ}(x,z)} \cdot \frac{f_{XZ}(x,z)}{f_X(x)} = \frac{f_{XYZ}(x,y,z)}{f_X(x)} = f_{YZ|X}(y,z|x)$$

The f_{XZ} terms cancel, which is the entire trick. So the bracket becomes

$$\int_z f_{YZ|X}(y,z|x)\,dz = f_{Y|X}(y|x)$$

because integrating z out of a joint conditional leaves the marginal conditional. Therefore

$$\mathbb{E}\Big\{\mathbb{E}[Y|X,Z] \;\Big|\; X\Big\} = \int_y y\, f_{Y|X}(y|x)\,dy = \mathbb{E}[Y|X=x] \quad \blacksquare$$

The idea in one sentence: conditioning on extra information and then averaging that information away returns you to the original conditional expectation.

19 Block 14. Mock midterm

60 minutes

Ten questions built on the exact blueprint: the same question types as the Sample Midterm and PS6, with different numbers so it is genuine practice. **Do this closed-book with only your cheat sheet**, timed at 60 minutes. Solutions follow.

Mock Midterm

1. If $X \sim N(3, 16)$ and $Y = -2X + 5$, what is the distribution of Y? Then find $\mathrm{P}(Y > 1)$.
2. Let X have pdf $f_X(x) = 3x^2$ for $0 < x < 1$, zero otherwise. Find $\mathbb{E}[X]$ and $\mathrm{Var}(X)$.
3. Let $X \sim \mathrm{Uniform}(0, 1)$ and let $Y = -3 \ln X$. Derive the pdf of Y and identify the distribution.
4. Let X_1, X_2 have joint pdf $f(x_1, x_2) = 4x_1x_2$ for $0 < x_1 < 1$, $0 < x_2 < 1$. Are X_1 and X_2 independent? Justify.
5. Let X and Y have joint pdf $f_{XY}(x, y) = \frac{2}{3}(x + 2y)$ for $0 < x < 1, 0 < y < 1$, zero otherwise.
   (a) Verify the joint pdf integrates to 1.
   (b) Find the marginal densities $f_X(x)$ and $f_Y(y)$.
   (c) Find $\mathbb{E}[X]$.
   (d) Are X and Y independent? Justify with a named theorem.
   (e) Find the conditional density $f_{Y|X}(y|x)$.
   (f) Find $\mathbb{E}[Y|X = x]$.
   (g) Find $\mathrm{P}\big(Y < \frac{1}{2} \mid X = \frac{1}{2}\big)$.
   (h) Use a result from class to evaluate $\mathbb{E}\big[\mathbb{E}[Y|X]\big]$.
6. Let $\mathrm{Var}(X) = 4$, $\mathrm{Var}(Y) = 9$, and $\mathrm{Cov}(X, Y) = 3$. Define $W = 2X - Y$ and $U = X + 3Y$. Find $\mathrm{Cov}(W, U)$, $\mathrm{Var}(W)$, $\mathrm{Var}(U)$, and $\mathrm{Corr}(W, U)$.
7. Show that if $X \sim \mathrm{Poisson}(\lambda)$ then $\mathrm{Var}(X) = \lambda$, using the mgf $M_X(t) = e^{\lambda(e^t - 1)}$.
8. A test for a condition has sensitivity $\mathrm{P}(+|D) = 0.95$ and specificity $\mathrm{P}(-|\neg D) = 0.90$. The condition affects 2% of the population. Find $\mathrm{P}(D|+)$.
9. Let X be a random variable with $\mathbb{E}[X] = 10$ and $\mathrm{Var}(X) = 4$. Give a bound on $\mathrm{P}(|X - 10| \geq 6)$ and name the inequality you used.
10. Let X be a discrete random variable with $\mathrm{P}(X = 1) = \frac{1}{2}$, $\mathrm{P}(X = 2) = \frac{1}{3}$, $\mathrm{P}(X = 4) = \frac{1}{6}$. Let $Y = \frac{1}{X}$. Find the pmf of Y, and state whether $\mathbb{E}[Y]$ equals $\frac{1}{\mathbb{E}[X]}$.

Mock midterm solutions

Solution 1

$$\mathbb{E}[Y] = -2(3) + 5 = -1, \qquad \mathrm{Var}(Y) = (-2)^2(16) = 64 \;\Rightarrow\; \boxed{Y \sim N(-1, 64)}$$

So $\sigma_Y = 8$. Standardize:

$$\mathrm{P}(Y > 1) = 1 - \mathrm{P}(Y \le 1) = 1 - \mathrm{P}\left(Z \le \frac{1-(-1)}{8}\right) = 1 - \Phi(0.25) = 1 - 0.5987 = \boxed{0.4013}$$

Note $a = -2$ is negative but $a^2 = 4$ is positive; variance can never be negative.

Solution 2

$$\mathbb{E}[X] = \int_0^1 x(3x^2)\,dx = \int_0^1 3x^3\,dx = \left[\tfrac{3x^4}{4}\right]_0^1 = \boxed{\tfrac{3}{4}}$$

$$\mathbb{E}[X^2] = \int_0^1 x^2(3x^2)\,dx = \int_0^1 3x^4\,dx = \left[\tfrac{3x^5}{5}\right]_0^1 = \tfrac{3}{5}$$

$$\mathrm{Var}(X) = \tfrac{3}{5} - \left(\tfrac{3}{4}\right)^2 = \tfrac{3}{5} - \tfrac{9}{16} = \tfrac{48-45}{80} = \boxed{\tfrac{3}{80} = 0.0375}$$

Solution 3

Here $f_X(x) = 1$ on $(0,1)$. Four steps.

1. **Support**: $x \to 0^+$ gives $y \to +\infty$; $x = 1$ gives $y = 0$. So $\mathcal{Y} = (0, \infty)$.
2. **Monotone**: $\dfrac{dy}{dx} = -\dfrac{3}{x} < 0$ on $(0,1)$. Strictly decreasing. ✓
3. **Invert**: $y = -3\ln x \Rightarrow -\frac{y}{3} = \ln x \Rightarrow x = e^{-y/3}$, and $\dfrac{dg^{-1}}{dy} = -\frac{1}{3}e^{-y/3}$.
4. **Apply**: $f_Y(y) = f_X(e^{-y/3}) \cdot \left|-\frac{1}{3}e^{-y/3}\right| = (1) \cdot \frac{1}{3}e^{-y/3}$.

$$\boxed{f_Y(y) = \tfrac{1}{3}e^{-y/3}, \quad y \in (0,\infty), \quad 0 \text{ otherwise}}$$

This is an exponential density with $\mathbb{E}[Y] = 3$, i.e. Exponential($\lambda = 3$) in the professor's convention. *Generalization worth noting: $-\lambda \ln U$ with U uniform gives an exponential with mean λ.*

Solution 4

$f(x_1, x_2) = 4x_1x_2 = (2x_1) \cdot (2x_2)$, a product of a function of x_1 alone and a function of x_2 alone, and the support $(0,1) \times (0,1)$ is a rectangle. By the **Factorization Theorem (Lemma 4.2.7)**, X_1 and X_2 are **independent**. *Confirmation: $f_{X_1}(x_1) = \int_0^1 4x_1x_2\,dx_2 = 2x_1$ and likewise $f_{X_2}(x_2) = 2x_2$, and their product is the joint.*

Solution 5, the big one

(a) Integrates to 1.

$$\int_0^1\int_0^1 \tfrac{2}{3}(x+2y)\,dx\,dy = \tfrac{2}{3}\int_0^1 \left[\tfrac{x^2}{2} + 2xy\right]_0^1 dy = \tfrac{2}{3}\int_0^1 \left(\tfrac{1}{2} + 2y\right) dy = \tfrac{2}{3}\left(\tfrac{1}{2} + 1\right) = \tfrac{2}{3}\cdot\tfrac{3}{2} = 1 \;✓$$

(b) Marginals. Integrate the other variable out, over $(0,1)$ each time.

$$f_X(x) = \int_0^1 \tfrac{2}{3}(x+2y)\,dy = \tfrac{2}{3}\Big[xy+y^2\Big]_0^1 = \boxed{\tfrac{2}{3}(x+1)}, \quad 0<x<1$$

$$f_Y(y) = \int_0^1 \tfrac{2}{3}(x+2y)\,dx = \tfrac{2}{3}\left[\tfrac{x^2}{2}+2xy\right]_0^1 = \tfrac{2}{3}\left(\tfrac{1}{2}+2y\right) = \boxed{\tfrac{1}{3}+\tfrac{4}{3}y}, \quad 0<y<1$$

(c) $\mathbb{E}[X]$.

$$\mathbb{E}[X] = \int_0^1 x\cdot\tfrac{2}{3}(x+1)\,dx = \tfrac{2}{3}\int_0^1 (x^2+x)\,dx = \tfrac{2}{3}\left(\tfrac{1}{3}+\tfrac{1}{2}\right) = \tfrac{2}{3}\cdot\tfrac{5}{6} = \boxed{\tfrac{5}{9}}$$

(d) Independence. $\frac{2}{3}(x+2y)$ is a bare sum of an x-term and a y-term and cannot be written as $g(x)h(y)$. By the **Factorization Theorem (Lemma 4.2.7)**, X and Y are **not independent**.

(e) Conditional density.

$$f_{Y|X}(y|x) = \frac{f_{XY}(x,y)}{f_X(x)} = \frac{\frac{2}{3}(x+2y)}{\frac{2}{3}(x+1)} = \boxed{\frac{x+2y}{x+1}}, \quad 0<y<1$$

Check it integrates to 1 over y: $\frac{1}{x+1}\int_0^1 (x+2y)\,dy = \frac{x+1}{x+1} = 1$ ✓

(f) Conditional expectation. Integrate over y, holding x fixed.

$$\mathbb{E}[Y|X=x] = \int_0^1 y\cdot\frac{x+2y}{x+1}\,dy = \frac{1}{x+1}\int_0^1 \left(xy+2y^2\right)\,dy = \frac{1}{x+1}\left[\frac{xy^2}{2}+\frac{2y^3}{3}\right]_0^1$$

$$= \boxed{\frac{1}{x+1}\left(\frac{x}{2}+\frac{2}{3}\right) = \frac{3x+4}{6(x+1)}}$$

(g) A conditional probability. Set $x=\frac{1}{2}$ in (e): $f_{Y|X}\left(y \mid \tfrac{1}{2}\right) = \dfrac{\frac{1}{2}+2y}{\frac{3}{2}} = \dfrac{1+4y}{3}$.

$$\mathrm{P}\left(Y<\tfrac{1}{2} \mid X=\tfrac{1}{2}\right) = \int_0^{1/2}\frac{1+4y}{3}\,dy = \frac{1}{3}\left[y+2y^2\right]_0^{1/2} = \frac{1}{3}\left(\tfrac{1}{2}+\tfrac{1}{2}\right) = \boxed{\tfrac{1}{3}}$$

(h) Iterated expectation. By the **Law of Iterated Expectations (Thm. 4.4.3)**, $\mathbb{E}\big[\mathbb{E}[Y|X]\big] = \mathbb{E}[Y]$. Compute $\mathbb{E}[Y]$ from the marginal in (b):

$$\mathbb{E}[Y] = \int_0^1 y\left(\tfrac{1}{3}+\tfrac{4}{3}y\right)\,dy = \tfrac{1}{3}\cdot\tfrac{1}{2}+\tfrac{4}{3}\cdot\tfrac{1}{3} = \tfrac{1}{6}+\tfrac{4}{9} = \tfrac{3}{18}+\tfrac{8}{18} = \boxed{\tfrac{11}{18}}$$

Verification (optional, and a great habit): $\int_0^1 \frac{3x+4}{6(x+1)}\cdot\frac{2}{3}(x+1)\,dx = \frac{1}{9}\int_0^1 (3x+4)\,dx = \frac{1}{9}\left(\frac{3}{2}+4\right) = \frac{11}{18}$ ✓

Solution 6

Covariance, by bilinearity (C4):

$$\mathrm{Cov}(2X-Y,\ X+3Y) = 2\mathrm{Cov}(X,X)+6\mathrm{Cov}(X,Y)-\mathrm{Cov}(Y,X)-3\mathrm{Cov}(Y,Y)$$

$$= 2(4)+6(3)-3-3(9) = 8+18-3-27 = \boxed{-4}$$

Variances, by (C5):

$$\text{Var}(W) = \text{Var}(2X - Y) = 4(4) + 1(9) + 2(2)(-1)(3) = 16 + 9 - 12 = \boxed{13}$$

$$\text{Var}(U) = \text{Var}(X + 3Y) = 1(4) + 9(9) + 2(1)(3)(3) = 4 + 81 + 18 = \boxed{103}$$

Correlation:

$$\text{Corr}(W, U) = \frac{-4}{\sqrt{13 \cdot 103}} = \frac{-4}{\sqrt{1339}} \approx \boxed{-0.109}$$

Check $|\rho| \leq 1$ ✓, and check the inputs were legitimate: $\rho_{XY} = 3/(2 \cdot 3) = 0.5$, valid.

Solution 7

$$M_X'(t) = e^{\lambda(e^t - 1)} \cdot \lambda e^t \;\Rightarrow\; \mathbb{E}[X] = M_X'(0) = e^0 \cdot \lambda = \lambda$$

For the second derivative, product rule on $e^{\lambda(e^t-1)}$ and λe^t:

$$M_X''(t) = e^{\lambda(e^t-1)} \left(\lambda e^t\right)^2 + e^{\lambda(e^t-1)} \lambda e^t \;\Rightarrow\; \mathbb{E}[X^2] = M_X''(0) = \lambda^2 + \lambda$$

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \lambda^2 + \lambda - \lambda^2 = \boxed{\lambda}$$

Mean equals variance, the signature property of the Poisson.

Solution 8

First derive the false positive rate from the *same* conditioning side:

$$\text{P}(+|\neg D) = 1 - \text{P}(-|\neg D) = 1 - 0.90 = 0.10$$

Then Bayes with the partition $\{D, \neg D\}$, where $\text{P}(D) = 0.02$ and $\text{P}(\neg D) = 0.98$:

$$\text{P}(D|+) = \frac{(0.95)(0.02)}{(0.95)(0.02) + (0.10)(0.98)} = \frac{0.019}{0.019 + 0.098} = \frac{0.019}{0.117} \approx \boxed{0.162}$$

Interpretation: even after a positive result on a 95% sensitive test, only about a 16% chance, because the base rate is low.

Solution 9

This asks about distance from the mean, so use **Chebyshev's inequality**:

$$\text{P}(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2} \;\Rightarrow\; \text{P}(|X - 10| \geq 6) \leq \frac{4}{36} = \boxed{\tfrac{1}{9} \approx 0.111}$$

Markov would not apply directly here: we are not told $X \geq 0$, and the question is about a two-sided deviation.

Solution 10

$Y = 1/X$ is strictly decreasing on the support $\{1, 2, 4\}$, and $g^{-1}(y) = 1/y$. Discrete case, so **no Jacobian**.

$$\mathcal{Y} = \left\{1, \tfrac{1}{2}, \tfrac{1}{4}\right\}, \qquad f_Y(y) = f_X\left(\tfrac{1}{y}\right)$$

$$\boxed{\mathrm{P}(Y=1)=\tfrac{1}{2},\quad \mathrm{P}\big(Y=\tfrac{1}{2}\big)=\tfrac{1}{3},\quad \mathrm{P}\big(Y=\tfrac{1}{4}\big)=\tfrac{1}{6}}$$

Second part. $\mathbb{E}[X]=1\cdot\frac{1}{2}+2\cdot\frac{1}{3}+4\cdot\frac{1}{6}=\frac{1}{2}+\frac{2}{3}+\frac{2}{3}=\frac{11}{6}$, so $\frac{1}{\mathbb{E}[X]}=\frac{6}{11}\approx 0.545$.

$$\mathbb{E}[Y]=1\cdot\tfrac{1}{2}+\tfrac{1}{2}\cdot\tfrac{1}{3}+\tfrac{1}{4}\cdot\tfrac{1}{6}=\tfrac{1}{2}+\tfrac{1}{6}+\tfrac{1}{24}=\tfrac{12+4+1}{24}=\tfrac{17}{24}\approx 0.708$$

They are not equal, illustrating $\mathbb{E}[g(X)]\neq g(\mathbb{E}[X])$. Since $g(x)=1/x$ is convex on $x>0$, **Jensen's inequality** predicts $\mathbb{E}[1/X]>1/\mathbb{E}[X]$, and indeed $0.708>0.545$. ✓

20 Bonus: four harder problems

You asked to be ready for slightly harder than the exam. These stretch the same machinery.

Bonus problems

B1. Let X and Y have joint pdf $f_{XY}(x,y) = 8xy$ for $0 < x < y < 1$, zero otherwise. Note the support is *triangular*. (a) Find $f_X(x)$ and $f_Y(y)$. (b) Are X and Y independent? (c) Find $\mathbb{E}[X|Y=y]$.

B2. Let (X,Y) be bivariate normal with $\mu_X = 2$, $\mu_Y = 5$, $\sigma_X^2 = 4$, $\sigma_Y^2 = 9$, $\rho = 0.6$. Find the full conditional distribution of Y given $X = 4$, and then find $\mathrm{P}(Y > 8 \mid X = 4)$.

B3. Let $X \sim \text{Uniform}(0,1)$ and define $Y = X^2$ and $W = 1 - X$. Find $\text{Corr}(X,W)$ without computing any integrals, and explain why.

B4. Suppose $N \sim \text{Poisson}(\lambda)$ and, given $N = n$, $Y \sim \text{Binomial}(n,p)$. Find $\mathbb{E}[Y]$ using the Law of Iterated Expectations.

Bonus solution B1: the non-rectangular support

This is the case Block 4 warned about. The region is the triangle $0 < x < y < 1$.

(a) Marginals. For $f_X(x)$, integrate out y. Given x, the constraint $x < y < 1$ sets the limits:

$$f_X(x) = \int_x^1 8xy\,dy = 8x\left[\tfrac{y^2}{2}\right]_x^1 = 4x(1-x^2), \qquad 0 < x < 1$$

For $f_Y(y)$, integrate out x, and given y the constraint is $0 < x < y$:

$$f_Y(y) = \int_0^y 8xy\,dx = 8y\left[\tfrac{x^2}{2}\right]_0^y = 4y^3, \qquad 0 < y < 1$$

The limits are not constants. That is the entire difficulty, and getting them from the inequality $0 < x < y < 1$ is the skill.

(b) Independence. The expression $8xy$ *looks* separable, but the support is a triangle, not a rectangle: knowing $X = x$ restricts Y to $(x,1)$. So the Factorization Theorem's condition fails, and X and Y are **not independent**. Confirm: $f_X(x)f_Y(y) = 4x(1-x^2)\cdot 4y^3 \neq 8xy$.

(c) Conditional expectation.

$$f_{X|Y}(x|y) = \frac{8xy}{4y^3} = \frac{2x}{y^2}, \qquad 0 < x < y$$

$$\mathbb{E}[X|Y=y] = \int_0^y x\cdot\frac{2x}{y^2}\,dx = \frac{2}{y^2}\left[\frac{x^3}{3}\right]_0^y = \frac{2y^3}{3y^2} = \boxed{\frac{2y}{3}}$$

Sensible: given the ceiling y, the average of X sits two thirds of the way up, because the density $2x/y^2$ tilts mass toward larger x.

Bonus solution B2: bivariate normal conditioning

Use Fact 3. With $\sigma_X = 2$, $\sigma_Y = 3$:

$$\mathbb{E}[Y|X=4] = \mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(x-\mu_X) = 5 + 0.6\cdot\frac{3}{2}(4-2) = 5 + 0.9(2) = 6.8$$

$$\text{Var}(Y|X=4) = \sigma_Y^2(1-\rho^2) = 9(1-0.36) = 9(0.64) = 5.76 \;\Rightarrow\; \sigma_{Y|X} = 2.4$$

$$\boxed{Y \mid X=4 \;\sim\; N(6.8,\; 5.76)}$$

Now standardize within the conditional distribution:

$$\text{P}(Y > 8 \,|\, X=4) = 1 - \Phi\left(\frac{8-6.8}{2.4}\right) = 1 - \Phi(0.5) = 1 - 0.6915 = \boxed{0.3085}$$

Note the conditional variance 5.76 is smaller than the marginal variance 9. Knowing X bought you information.

Bonus solution B3: correlation with no integration

$W = 1 - X$ is an exactly linear function of X with negative slope. Recall the property: $\rho_{XY} = -1$ whenever $Y = aX + b$ with $a < 0$. Here $a = -1$, so

$$\boxed{\text{Corr}(X, W) = -1}$$

Verification by formula, no integrals needed:

$$\text{Cov}(X, 1-X) = \text{Cov}(X,1) - \text{Cov}(X,X) = 0 - \text{Var}(X) = -\text{Var}(X)$$

$$\text{Var}(W) = \text{Var}(1-X) = (-1)^2\text{Var}(X) = \text{Var}(X)$$

$$\rho = \frac{-\text{Var}(X)}{\sqrt{\text{Var}(X)}\sqrt{\text{Var}(X)}} = \frac{-\text{Var}(X)}{\text{Var}(X)} = -1$$

The distribution of X never entered. Perfect linear dependence gives $|\rho| = 1$ regardless of the underlying distribution, which is the real lesson here.

Bonus solution B4: iterated expectations with a random parameter

The number of trials is itself random. Condition on it. **Inner step:** given $N = n$, $Y \sim \text{Binomial}(n, p)$, so

$$\mathbb{E}[Y|N=n] = np \quad \Rightarrow \quad \mathbb{E}[Y|N] = Np$$

Outer step: apply the Law of Iterated Expectations, averaging over N:

$$\mathbb{E}[Y] = \mathbb{E}_N\big[\mathbb{E}[Y|N]\big] = \mathbb{E}[Np] = p\,\mathbb{E}[N] = \boxed{p\lambda}$$

This two-line argument replaces a genuinely painful double sum. It is exactly why LIE is on the exam, and it is the same move as PS6 #2(j). Aside for the curious: one can show $Y \sim Poisson(\lambda p)$ exactly, which is consistent with the mean we just found.

Two pages. Everything.

The next two pages are designed to be copied onto one double-sided sheet. Transcribe them by hand: the copying is a third of the studying, and you will find you can compress further once you understand it.

Page 1: the machines. Page 2: the reference tables.

GSE 518 MIDTERM CHEAT SHEET • SIDE 1: THE MACHINES

0. SYMBOL DECODER

X random variable (the machine) | x a value it takes | $\mathcal{X}$ the **support**, where $f_X > 0$ | $\mu = \mathbb{E}[X]$ center | $\sigma^2 = \mathrm{Var}(X)$ spread^2 | $\sigma = \sqrt{\sigma^2}$ | ρ correlation, in $[-1,1]$ | Φ standard normal *cdf*, ϕ its *pdf* | $g^{-1}(y)$: solve $y = g(x)$ for x. **In $N(\mu, \sigma^2)$ the second slot is the VARIANCE. Standardize by σ.**

1. CDF / PMF / PDF

$F_X(x) = \mathrm{P}(X \le x)$. $f_X(x) = \frac{\mathrm{d}F_X}{\mathrm{d}x}$, $F_X(x) = \int_{-\infty}^{x} f_X(r)\,\mathrm{d}r$.
Valid pdf/pmf: (i) $f \ge 0$; (ii) integrates/sums to 1.
Valid cdf: (i) limits 0 and 1; (ii) non-decreasing; (iii) right continuous.
Discrete: $f_X(x) = \mathrm{P}(X = x)$, jump height of F. Continuous: f is NOT a probability, can exceed 1; $\mathrm{P}(X = x) = 0$.
Write cdfs piecewise on all of $\mathbb{R}$, including the 0 and 1 branches.

2. EXPECTATION AND VARIANCE

$$\mathbb{E}[g(X)] = \textstyle\sum_x g(x) f_X(x) \ \text{ or } \ \int_{\mathcal{X}} g(x) f_X(x)\,\mathrm{d}x$$

$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$; $\mathbb{E}[c] = c$; $\mathbb{E}[g(X)] \ne g(\mathbb{E}[X])$.

$$\boxed{\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2} \qquad \mathrm{Var}(aX + b) = a^2\mathrm{Var}(X)$$

If the integral diverges, the moment **does not exist**. Say so.

3. TRANSFORMATIONS $Y = g(X)$ (4 STEPS)

(1) Find $\mathcal{Y}$ by pushing endpoints of $\mathcal{X}$ through g. **(2)** Check g strictly monotone *on* $\mathcal{X}$ via sign of $\mathrm{d}g/\,\mathrm{d}x$. **(3)** Invert: $x = g^{-1}(y)$; differentiate. **(4)** Apply:

$$\boxed{f_Y(y) = f_X(g^{-1}(y)) \left| \frac{\mathrm{d}g^{-1}(y)}{\mathrm{d}y} \right| \text{ on } \mathcal{Y},\ 0 \text{ o/w}}$$

DISCRETE: same but NO Jacobian: $f_Y(y) = f_X(g^{-1}(y))$.
cdf route: $g \uparrow$: $F_Y(y) = F_X(g^{-1}(y))$; $g \downarrow$: $F_Y(y) = 1 - F_X(g^{-1}(y))$.
Never forget: absolute value + state the support.

4. NORMAL SHORTCUTS

$$X \sim N(\mu, \sigma^2),\ Y = aX + b \ \Rightarrow\ \boxed{Y \sim N(a\mu + b,\ a^2\sigma^2)}$$

$$Z = \frac{X - \mu}{\sigma} \sim N(0,1); \quad \mathrm{P}(X \le c) = \Phi\left(\frac{c - \mu}{\sigma}\right)$$

$\Phi(-z) = 1 - \Phi(z)$; $\phi(-z) = \phi(z)$; $\Phi(0) = 0.5$; $Z^2 \sim \chi^2_{(1)}$.
No closed form for the normal cdf; use the table.

5. JOINT DISTRIBUTIONS (HALF THE EXAM)

Marginal: $f_X(x) = \int_{\mathcal{Y}} f_{XY}(x,y)\,\mathrm{d}y$ (integrate out the OTHER)

Conditional: $f_{Y|X}(y|x) = \dfrac{f_{XY}(x,y)}{f_X(x)}$ (joint ÷ marginal of the GIVEN one)

Cond. exp.: $\mathbb{E}[Y|X = x] = \int_{\mathcal{Y}} y\, f_{Y|X}(y|x)\,\mathrm{d}y$ (a function of x)

$$f_{XY} = \frac{\partial^2 F_{XY}}{\partial x \partial y}; \quad \mathbb{E}[g(X,Y)] = \iint g(x,y) f_{XY}\,\mathrm{d}x\,\mathrm{d}y$$

Range belongs to the variable LEFT of the bar.
Ask every time: which variable is random? Integrate over that one.

6. INDEPENDENCE

Factorization Thm (Lemma 4.2.7): $X \perp\!\!\!\perp Y \iff f_{XY}(x,y) = g(x)h(y)$.
Support must be a RECTANGLE. If limits like $0 < x < y < 1$, NOT independent even if it looks separable.
Sum $(x + y)$ does not factor. Product (xy) does. $e^{-(x+y)} = e^{-x}e^{-y}$ DOES.
If independent: $f_{Y|X} = f_Y$; $\mathbb{E}[r(X)s(Y)] = \mathbb{E}[r(X)]\mathbb{E}[s(Y)]$; $\mathrm{Cov} = 0$; $M_{X+Y} = M_X M_Y$; $g(X) \perp\!\!\!\perp h(Y)$.
$\mathrm{Cov} = 0$ **does NOT imply independence** (only linear dependence is measured). **Exception: bivariate normal, where $\rho = 0 \iff$ independent.**

7. COVARIANCE AND CORRELATION

$$\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]; \qquad \rho_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y}$$

$\mathrm{Cov}(X,X) = \mathrm{Var}(X)$; $\mathrm{Cov}(X,a) = 0$; Cov is symmetric.

$$\boxed{\mathrm{Cov}(aX + bY, cU + dV) = ac\mathrm{Cov}(X,U) + ad\mathrm{Cov}(X,V)}$$

$$\boxed{+bc\mathrm{Cov}(Y,U) + bd\mathrm{Cov}(Y,V)}$$

$$\boxed{\mathrm{Var}(aX + bY) = a^2\mathrm{Var}(X) + b^2\mathrm{Var}(Y) + 2ab\mathrm{Cov}(X,Y)}$$

Independent $\Rightarrow$ drop the cross term. **Expand like FOIL.**
$\rho = \pm 1 \iff Y = aX + b$ exactly (+ if $a > 0$). ρ is unitless. **If $|\rho| > 1$ you made an error.**

8. LIE AND JENSEN

$$\boxed{\mathbb{E}[Y] = \mathbb{E}_X[\mathbb{E}[Y|X]]} \qquad \mathbb{E}[r(Y)] = \mathbb{E}_X[\mathbb{E}[r(Y)|X]]$$

$$\mathbb{E}[Y|X] = \mathbb{E}_{Z|X}[\mathbb{E}[Y|X,Z] \mid X]$$

Outer expectation uses the marginal of whatever was RIGHT of the bar.
Jensen: g convex $\Rightarrow \mathbb{E}[g(X)] \ge g(\mathbb{E}[X])$; concave $\Rightarrow \le$.
$\mathbb{E}[X^2] > (\mathbb{E}[X])^2$; $\mathbb{E}[\ln X] < \ln \mathbb{E}[X]$; $\mathbb{E}[1/X] > 1/\mathbb{E}[X]$.

9. BIVARIATE NORMAL $N(\mu_X, \mu_Y, \sigma_X^2, \sigma_Y^2, \rho)$

Marginals normal. $\rho = 0 \iff$ independent.

$$Y|X \sim N\left(\mu_Y + \rho\tfrac{\sigma_Y}{\sigma_X}(X - \mu_X),\ \sigma_Y^2(1 - \rho^2)\right)$$

$\mathbb{E}[Y|X] = a + bX$ is LINEAR; slope $b = \rho\frac{\sigma_Y}{\sigma_X} = \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(X)}$ **= OLS β.**

$$aX + bY \sim N\left(a\mu_X + b\mu_Y,\ a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\rho\sigma_X\sigma_Y\right)$$

10. PROBABILITY AND BAYES

$\mathrm{P}(A^c) = 1 - \mathrm{P}(A)$; $\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$;
$A \subseteq B \Rightarrow \mathrm{P}(A) \le \mathrm{P}(B)$; DeMorgan $(A \cup B)^c = A^c \cap B^c$.
Boole: $\mathrm{P}(\cup A_i) \le \sum \mathrm{P}(A_i)$. Bonferroni: $\mathrm{P}(\cap A_i) \ge 1 - \sum[1 - \mathrm{P}(A_i)]$.

$$\mathrm{P}(A|B) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)}; \quad \mathrm{P}(A \cap B) = \mathrm{P}(B)\mathrm{P}(A|B)$$

Total prob: $\mathrm{P}(B) = \sum_i \mathrm{P}(B|A_i)\mathrm{P}(A_i)$

$$\boxed{\textbf{Bayes: } \mathrm{P}(A_i|B) = \frac{\mathrm{P}(B|A_i)\mathrm{P}(A_i)}{\sum_j \mathrm{P}(B|A_j)\mathrm{P}(A_j)}}$$

$\mathrm{P}(A|B) + \mathrm{P}(A^c|B) = 1$, but $\mathrm{P}(A|B) + \mathrm{P}(A|B^c) \ne 1$.
Independent events: $\mathrm{P}(A \cap B) = \mathrm{P}(A)\mathrm{P}(B)$. For 3+, need ALL subsets; pairwise $\ne$ mutual.

11. INEQUALITIES

Markov ($X \ge 0, c > 0$): $\mathrm{P}(X \ge c) \le \mathbb{E}[X]/c$.
Chebyshev (any $X, k > 0$): $\mathrm{P}(|X - \mu| \ge k) \le \sigma^2/k^2$.
With $k = m\sigma$: $\mathrm{P}(|X - \mu| \ge m\sigma) \le 1/m^2$.

12. DISTRIBUTIONS: $\mathbb{E}[X]$, $\mathrm{Var}(X)$, mgf

Dist	$\mathbb{E}[X]$	$\mathrm{Var}(X)$	$M_X(t)$
Bernoulli(p)	p	$p(1-p)$	$(1-p)+pe^t$
Binomial(n,p)	np	$np(1-p)$	$[(1-p)+pe^t]^n$
Geometric(p)	$\frac{1}{p}$	$\frac{1-p}{p^2}$	
NegBin(r,p)	$\frac{r(1-p)}{p}$	$\frac{r(1-p)}{p^2}$	
Poisson(λ)	λ	λ	$e^{\lambda(e^t-1)}$
Disc.Unif$\{1..N\}$	$\frac{N+1}{2}$	$\frac{N^2-1}{12}$	
Uniform(a,b)	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\frac{e^{tb}-e^{ta}}{t(b-a)}$
Normal(μ,σ^2)	μ	σ^2	$e^{\mu t+\sigma^2t^2/2}$
Gamma(α,β)	$\alpha\beta$	$\alpha\beta^2$	$(1-\beta t)^{-\alpha}$
Expon(λ)	λ	λ^2	$(1-\lambda t)^{-1}$
$\chi^2_{(p)}$	p	$2p$	$(1-2t)^{-p/2}$
Beta(α,β)	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	
$t_{(r)}$	0 $(r>1)$	$\frac{r}{r-2}$ $(r>2)$	none

pdfs: Bern $p^x(1-p)^{1-x}$ | Bin $\binom{n}{x}p^x(1-p)^{n-x}$ | Geom $p(1-p)^{x-1}$, $x\geq 1$ | Poisson $\frac{e^{-\lambda}\lambda^x}{x!}$ | Unif $\frac{1}{b-a}$ | Normal $\frac{1}{\sqrt{2\pi\sigma^2}}e^{-(x-\mu)^2/2\sigma^2}$ | Gamma $\frac{x^{\alpha-1}e^{-x/\beta}}{\Gamma(\alpha)\beta^\alpha}$ | $\chi^2_{(p)}$ $\frac{x^{p/2-1}e^{-x/2}}{\Gamma(p/2)2^{p/2}}$ | Beta $\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}$.

Prof's exponential convention: $f=\frac{1}{\lambda}e^{-x/\lambda}$, $\mathbb{E}=\lambda$. If given ce^{-cx}, call it "exponential with mean $1/c$".

Identities: Gamma$(1,\beta)$ =Exp(β); Gamma$(\frac{p}{2},2)$ = $\chi^2_{(p)}$; Beta$(1,1)$ =Unif$(0,1)$; Bin$(1,p)$ =Bern(p); NB$(1,p)$ =Geom(p); $Z^2\sim\chi^2_{(1)}$; $\sum\chi^2_{(p_i)}\sim\chi^2_{(\sum p_i)}$; $F(1,r)=t^2_{(r)}$.

13. MGF TOOLKIT

$$M_X(t)=\mathbb{E}[e^{tX}];\qquad \mathbb{E}[X^n]=M_X^{(n)}(0)$$

Thm 2.3.11: $M_X=M_Y$ near $0\Rightarrow X\sim Y$. **Cite it by name.**
Thm 4.2.12: $X\perp\!\!\!\perp Y\Rightarrow M_{X+Y}=M_X\cdot M_Y$.
Uniform mgf undefined at $t=0$, so it cannot generate moments.
Trick: to kill a hard integral, reshape the integrand into a known pdf and set the integral to 1.

14. $\Phi(z)$ TABLE (standard normal cdf)

z	$\Phi(z)$	z	$\Phi(z)$	z	$\Phi(z)$
0.00	.5000	0.90	.8159	1.80	.9641
0.10	.5398	1.00	.8413	1.90	.9713
0.25	.5987	1.10	.8643	1.96	.9750
0.50	.6915	1.20	.8849	2.00	.9772
0.60	.7257	1.28	.8997	2.33	.9901
0.675	.7500	1.40	.9192	2.50	.9938
0.75	.7734	1.645	.9500	2.58	.9951
0.80	.7881	1.75	.9599	3.00	.9987

$\Phi(-z)=1-\Phi(z)$. Central intervals: $\pm1\sigma\approx 68\%$, $\pm1.645\sigma=90\%$, $\pm1.96\sigma=95\%$, $\pm2.58\sigma=99\%$.

15. CALCULUS AND SERIES FACTS

$$\int x^n\,dx=\tfrac{x^{n+1}}{n+1};\quad \int\tfrac{1}{x}\,dx=\ln x;\quad \int e^{ax}\,dx=\tfrac{e^{ax}}{a}$$

$$\int_0^\infty e^{-ax}\,dx=\tfrac{1}{a};\quad \int_0^\infty xe^{-ax}\,dx=\tfrac{1}{a^2};\quad \int_0^\infty u^{\alpha-1}e^{-u}\,du=\Gamma(\alpha)$$

$$\Gamma(1)=1,\Gamma(\tfrac{1}{2})=\sqrt{\pi},\Gamma(\alpha+1)=\alpha\Gamma(\alpha),\Gamma(n)=(n-1)!$$

$$\sum_{i=1}^n i=\tfrac{n(n+1)}{2};\quad \sum_{i=1}^n i^2=\tfrac{n(n+1)(2n+1)}{6};\quad \sum_{i=0}^\infty ar^i=\tfrac{a}{1-r}\ (|r|<1)$$

$\binom{n}{x}=\frac{n!}{(n-x)!\,x!}$. Double integral: do the INNER variable first, treating the outer as constant.

16. THE PLAYBOOK: SIGNAL → MOVE

Sees	Does
"find c so it's a pdf"	set $\int f=1$, solve
"find the cdf"	$\int_{-\infty}^x f$; piecewise, all of $\mathbb{R}$
"$Y=g(X)$, find f_Y"	4-step change of variable
"$X\sim N$, $Y=aX+b$"	$N(a\mu+b,a^2\sigma^2)$
"$P(X\leq c)$", normal	standardize, $\Phi\left(\frac{c-\mu}{\sigma}\right)$
"find $\mathrm{Var}(X)$"	$\mathbb{E}[X^2]-(\mathbb{E}[X])^2$
"find the marginal"	integrate out the other one
"independent?"	Factorization + rectangle check
"find $f_{Y\|X}$"	joint ÷ marginal of X
"find $\mathbb{E}[Y\|X=x]$"	$\int yf_{Y\|X}\,dy$, over y
"$\mathbb{E}[\mathbb{E}[Y\|X]]$"	LIE $=\mathbb{E}[Y]$, do not integrate
"Corr(W,U), linear combos"	FOIL the Cov, then $\mathrm{Var}(aX+bY)$
"show $X\sim$ some dist"	mgf, match, cite Thm 2.3.11
"bound a probability"	Markov $(X\geq 0)$ or Chebyshev
"$\mathbb{E}[g(X)]$ vs $g(\mathbb{E}[X])$"	Jensen
"given $P(B\|A)$, want $P(A\|B)$"	Bayes
"a partition, then event B"	Law of Total Probability

17. DEDUCTION-KILLERS (check before submitting)

1. State the support on every density. **2.** cdf needs 0 and 1 branches. **3.** Absolute value in the Jacobian. **4.** Conditional's range belongs to the LEFT-of-bar variable. **5.** Standardize by σ, not σ^2. **6.** Marginal of X integrates out y. **7.** Never say uncorrelated $\Rightarrow$ independent (unless bivariate normal). **8.** NAME the theorem. **9.** If it says "use a result from class", do not integrate. **10.** Discrete: $P(X\geq 2)=1-P(X\leq 1)$. **11.** Check densities integrate to 1. **12. Show all work: setup earns more than the answer.**

18. THE PS6#2 / SM#5 TEMPLATE (do this in order)

Given f_{XY} on a rectangle: **(1)** write the support. **(2)** check $\iint f=1$. **(3)** $f_X=\int_{\mathcal{Y}}f\,dy$, $f_Y=\int_{\mathcal{X}}f\,dx$. **(4)** independence via factorization. **(5)** $f_{Y|X}=f_{XY}/f_X$, state y's range. **(6)** $\mathbb{E}[Y|X=x]=\int yf_{Y|X}\,dy$. **(7)** conditional probability: plug in the given value, integrate. **(8)** $\mathbb{E}[\mathbb{E}[Y|X]]=\mathbb{E}[Y]$ by LIE. **(9)** for ρ: need $\mathbb{E}X$, $\mathbb{E}Y$, $\mathbb{E}X^2$, $\mathbb{E}Y^2$, $\mathbb{E}[XY]=\iint xyf$; then $\mathrm{Cov}=\mathbb{E}[XY]-\mathbb{E}X\mathbb{E}Y$ and divide by $\sigma_X\sigma_Y$.

Known answers: PS6#2 $(0.25(2x+y),\ 0<x<1,\ 0<y<2)$: $f_X=x+\frac{1}{2}$, $f_Y=\frac{1+y}{4}$, $\mathbb{E}X=\frac{7}{12}$, $\mathbb{E}[X|Y=y]=\frac{1}{1+y}(\frac{2}{3}+\frac{y}{2})$. SM#5 ($x+y$ on unit square): $f_X=x+\frac{1}{2}$, $\mathbb{E}X=\mathbb{E}Y=\frac{7}{12}$, $\mathrm{Var}=\frac{11}{144}$, $\mathbb{E}[XY]=\frac{1}{3}$, $\mathrm{Cov}=-\frac{1}{144}$, $\rho=-\frac{1}{11}$, $\mathbb{E}[Y|X]=\frac{3x+2}{6x+3}$.